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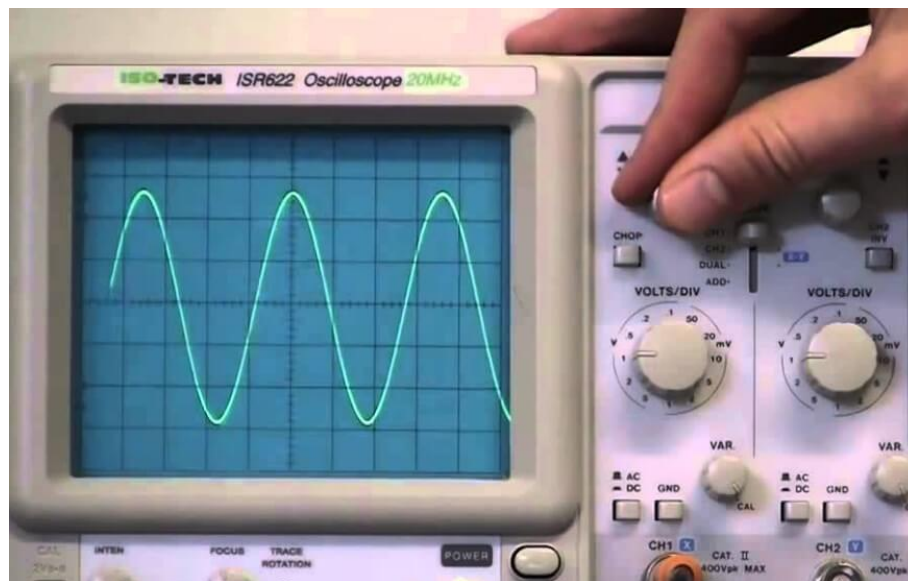
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*Electronic Instruments and Measurements for
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Group composed by:

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Exercise 1:

Using the LabVIEW software, create the following VIs:

1. Generate a sinusoidal signal at $f=1\text{kHz}$ of amplitude 1V and, after an appropriate and discussed evaluation of the sampling parameters, show, through one or more graphs, the signal in both the time and frequency domains. Highlight whether the sampling performed is synchronous or asynchronous and motivate its response by calculating the number of signal periods contained in the chosen observation window. Choose a window function from those proposed by the software (Hanning, Hamming, etc.) and, after justifying its choice, measure the difference between the amplitude measured with the rectangular window and the one chosen.
 - a. Set the sampling parameters to be in any asynchronous sampling condition. Apply zero-padding to see if a reduction in scallop loss occurs. Choose an arbitrary number of zeros until a minimum error condition is reached.
2. Verify the convolution theorem, that is, verify that:
 - $FT[f * g] = FT[f] \cdot FT[g]$
 - $FT[f \cdot g] = FT[f] * FT[g]$

Generate a square wave signal at 2 kHz. Set the sampling parameters to be aliased. Once a sampling rate has been chosen, perform a calculation (with any software) to predict the visible aliasing frequencies in the spectrum between 0 and $f_s/2$. Check in the frequency domain what was calculated.



Methodology

1. Signal Generation

- Generate a sinusoidal signal at **1 kHz** with an amplitude of **1 V** using LabVIEW.

2. Sampling Parameters

- **Sampling Frequency (F_s): 10 kHz**
- **Number of Samples (N): 995**
- **Observation Time:**
- The chosen sampling frequency results in an **asynchronous sampling process**, because the observation window does not contain an integer number of signal periods.

3. Time and Frequency Domain Analysis

- Display the generated signal on graphs to visualize both the time and frequency domains.
- **Time Domain:** Plot the sinusoidal waveform over the observation window.
- **Frequency Domain:** Apply the Fourier Transform (FFT) and display the frequency spectrum of the signal.

4. Sampling Analysis

- Signal frequency:
 $f=1000$ Hz
- Signal period:
- Number of signal periods contained in the observation window:
- Therefore, the observation window contains **99.5 periods** of the sinusoidal signal.
- Since the number of periods is **not an integer**, the sampling is **asynchronous**.
- This condition produces **spectral leakage** in the frequency domain when a rectangular window is used.

5. Window Function

- **Choice:** Hanning Window
- **Justification:**

The Hanning window is selected to reduce spectral leakage caused by the asynchronous sampling condition and to improve the frequency-domain representation of the signal.

- **Comparison:**
 - Analyze the signal using both the **Rectangular Window** and the **Hanning Window**.



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- Compare the FFT results obtained with the two windows.
- Observe that the Hanning window reduces spectral leakage and concentrates the spectral energy around the fundamental frequency of 1 kHz.

6. Results Visualization

- Display the generated sinusoidal signal in the **time domain**.
- Display the FFT spectrum in the **frequency domain**.
- Compare the frequency spectra obtained with:
 - Rectangular Window
 - Hanning Window

The comparison demonstrates that the **Hanning window significantly reduces spectral leakage**, providing a cleaner frequency spectrum than the rectangular window under asynchronous sampling conditions. Since the observation window contains **99.5 signal periods**, the FFT does not perfectly align with the signal frequency bins, making the use of a window function particularly important. The results highlight the influence of sampling parameters and window selection on signal analysis in LabVIEW.



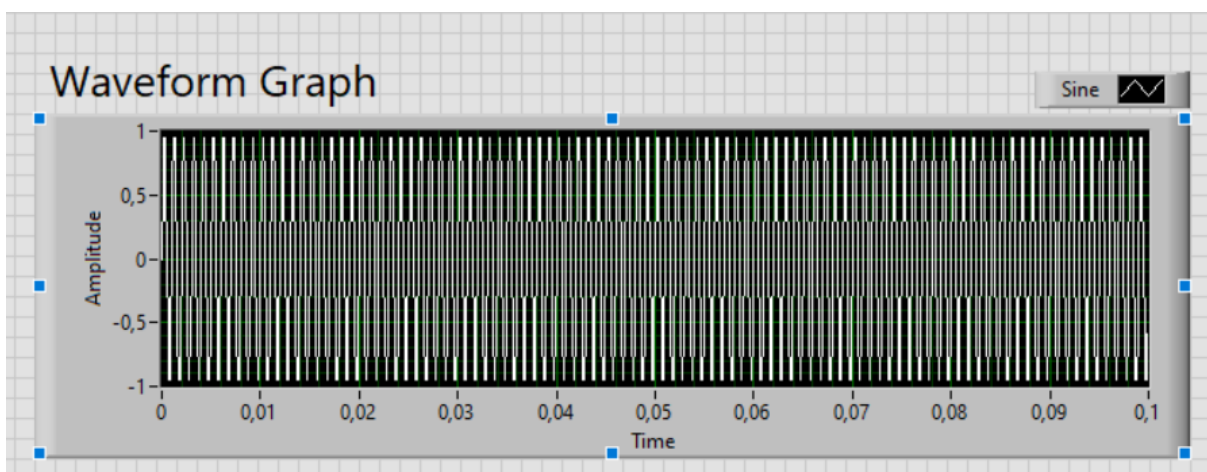
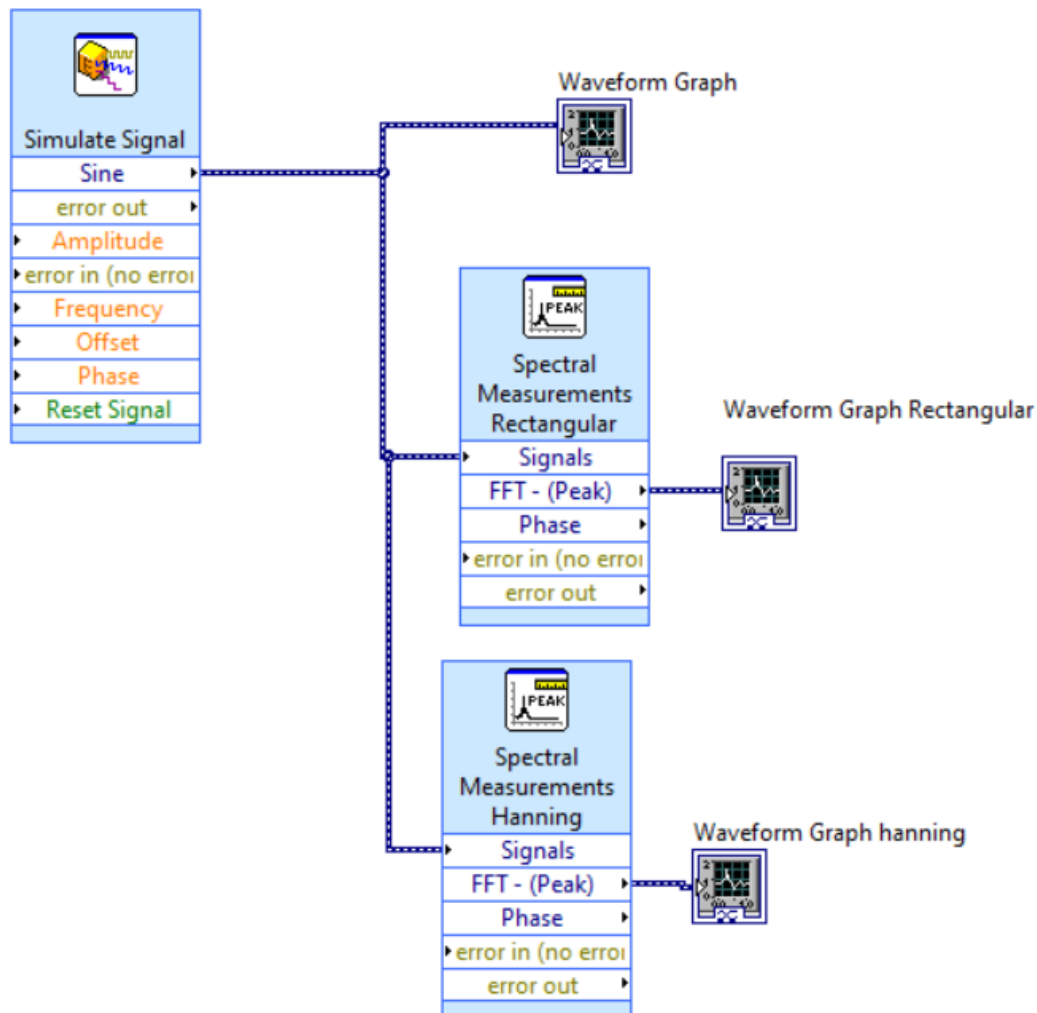
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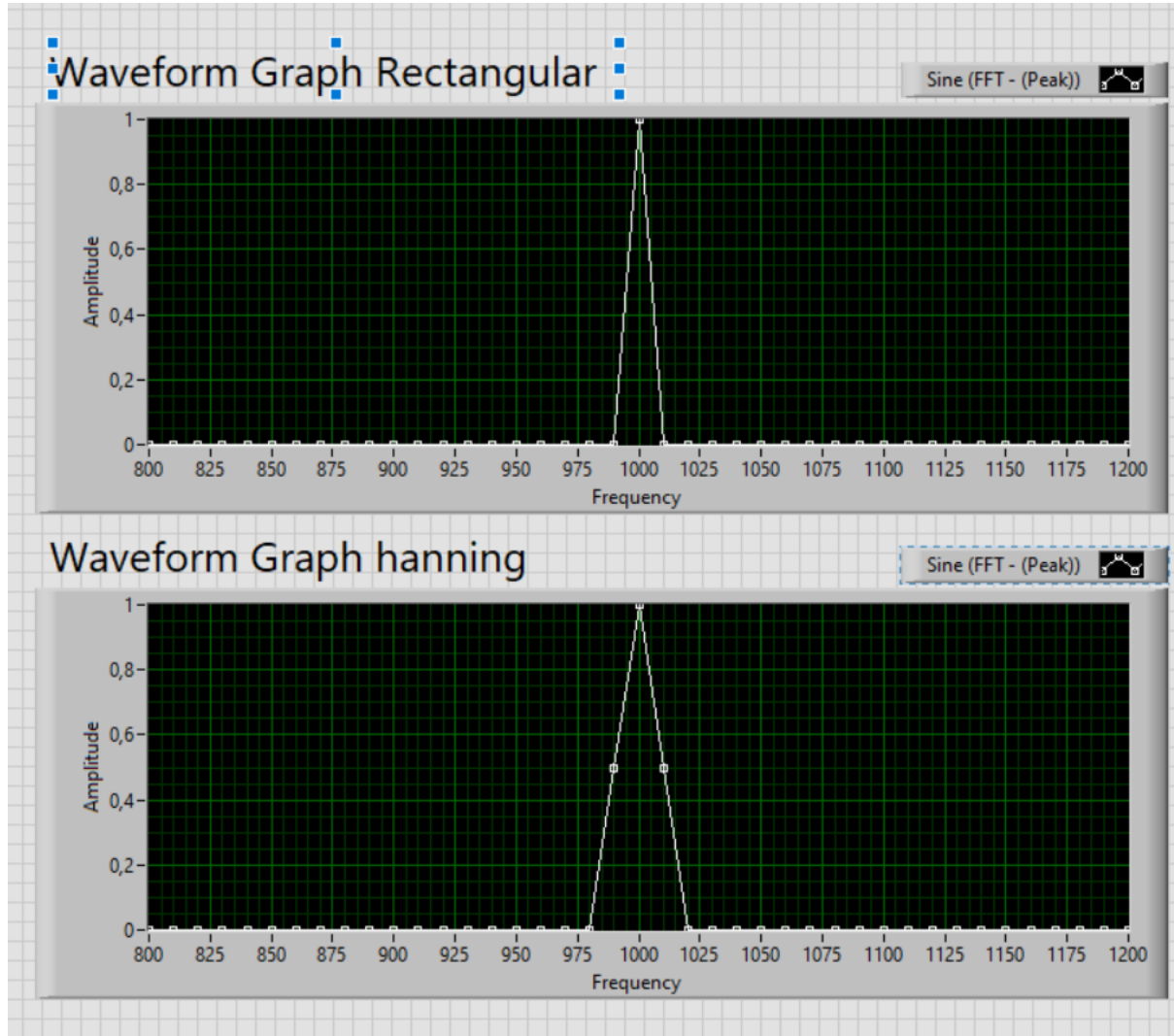
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Task 2: Reduction of Scallop Loss Using Zero-Padding

Objective

- To reduce scallop loss under asynchronous sampling conditions by applying zero-padding and improving spectral accuracy.

Methodology

- An asynchronous sampling condition was created using:
 - Signal Frequency: **250 Hz**
 - Sampling Frequency: **790 Hz**
 - Window Length: **293 samples**
- Zero-padding was applied to the sampled signal to increase the FFT length.
- The number of added zeros was gradually increased while monitoring the frequency spectrum.
- The spectral response was evaluated until the minimum error condition and the best frequency representation were achieved.

Results

- Zero-padding improved the localization of the spectral peak.
- Scallop loss was significantly reduced compared to the original FFT result.
- The frequency-domain representation became more accurate and easier to interpret.
- Comparison of the spectra with and without zero-padding confirmed an improvement in spectral resolution and frequency estimation accuracy.



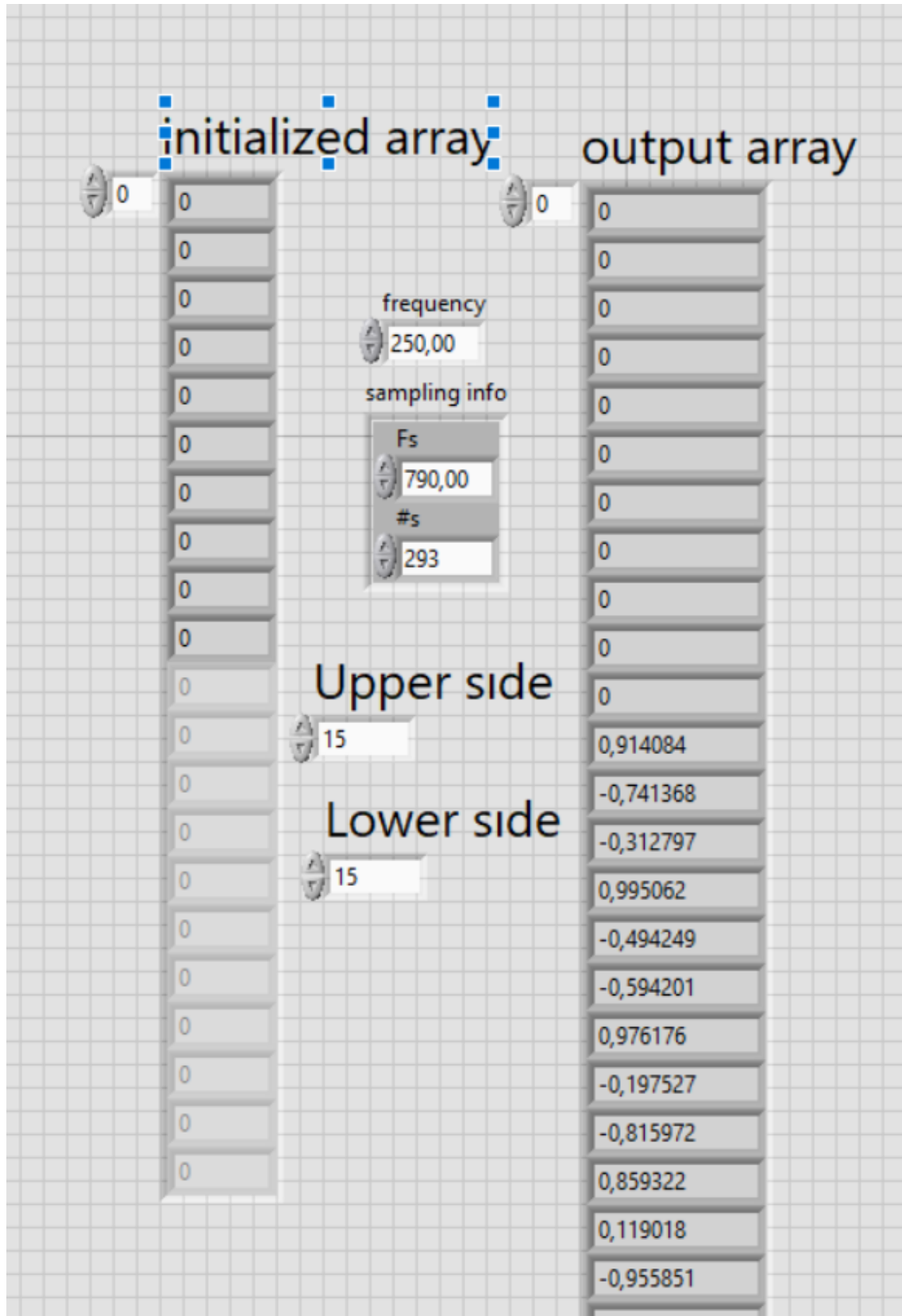
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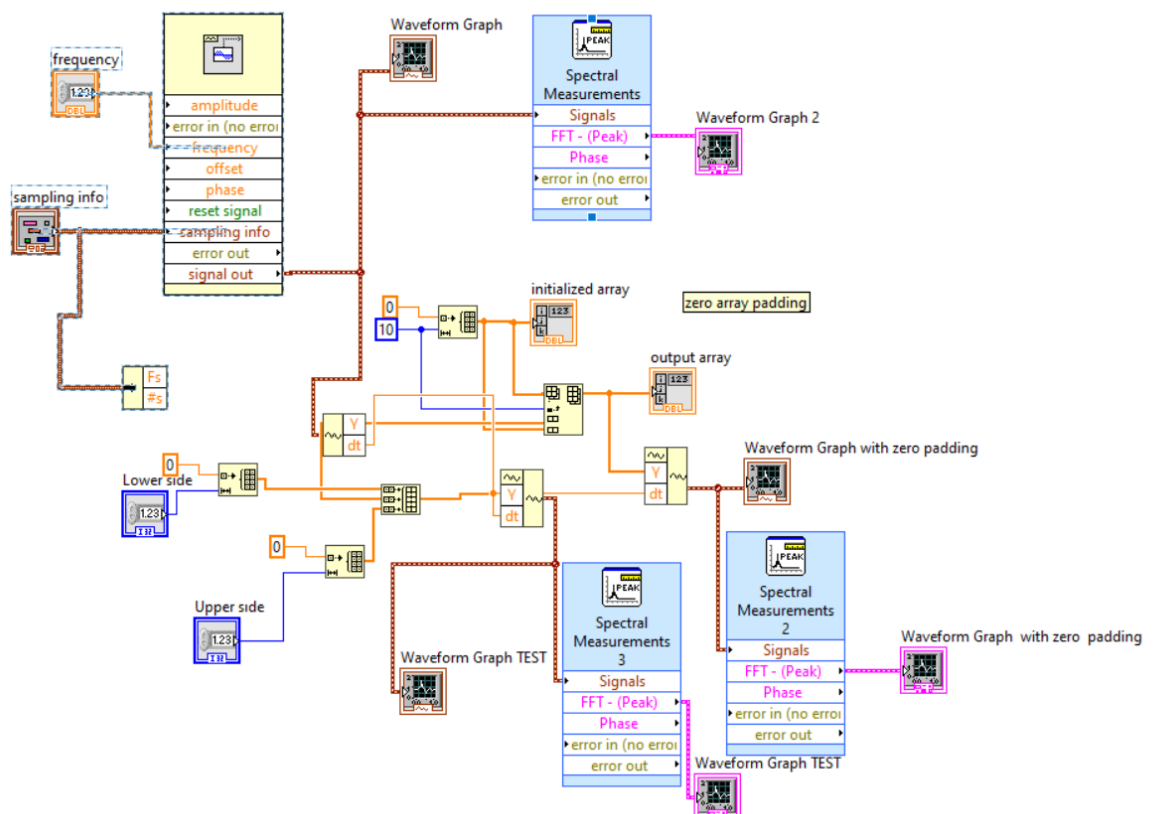
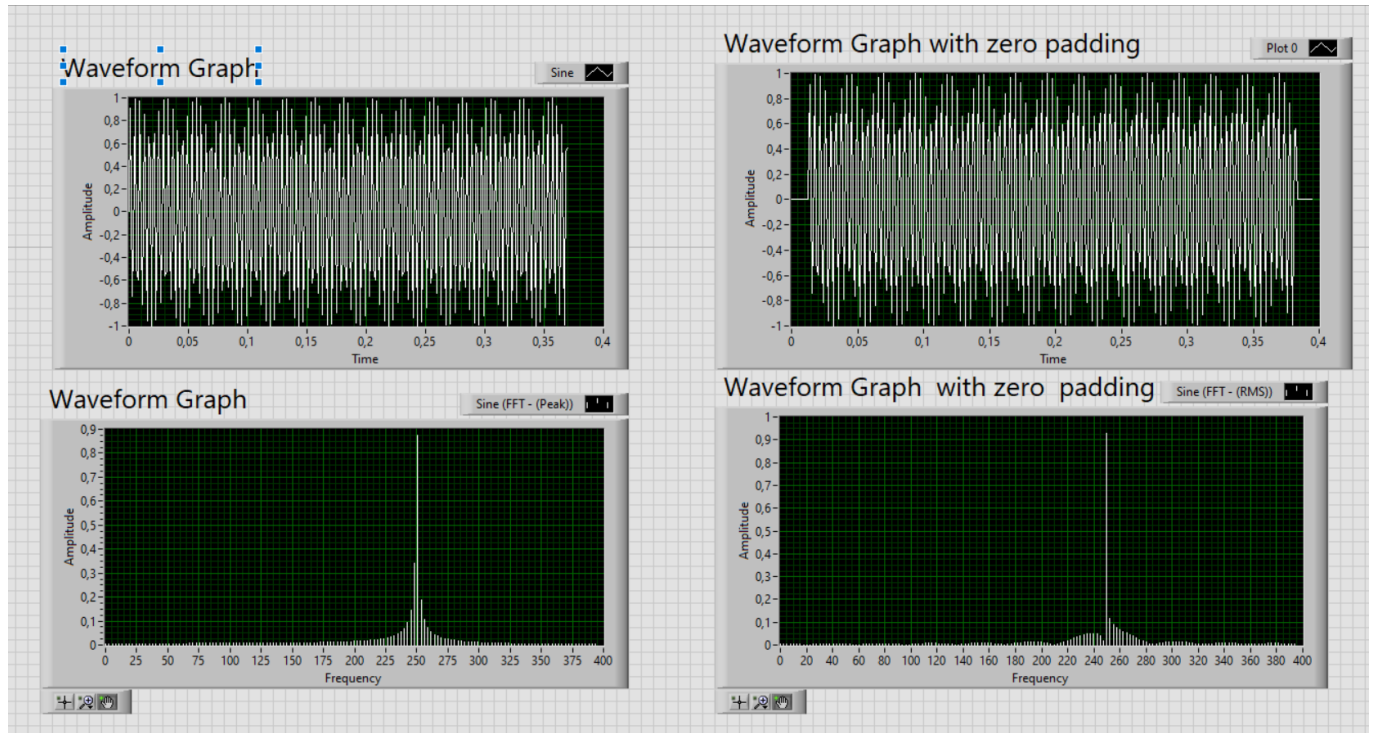
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Task 3: Verification of the Convolution Theorem Using LabVIEW

*Part 1: Verification of $FT[f * g] = FT[f] \cdot FT[g]$*

Methodology

- Two arbitrary signals, **f(t)** and **g(t)**, were generated in LabVIEW.
- The signals were convolved in the time domain to obtain **f(t) * g(t)**.
- Fourier Transforms (FT) of both signals were calculated, yielding **FT[f]** and **FT[g]**.
- The transformed signals were multiplied in the frequency domain.
- An Inverse Fourier Transform (IFT) was applied to the multiplication result.
- The obtained signal was compared with the direct time-domain convolution result.

Results

- The frequency-domain multiplication followed by IFT produced a result that closely matched the direct time-domain convolution.
- This verified the convolution theorem:

*Part 2: Verification of $FT[f \cdot g] = FT[f] * FT[g]$*

Methodology

- Two arbitrary signals, **f(t)** and **g(t)**, were generated in LabVIEW.
- The signals were multiplied directly in the time domain to obtain **f(t) · g(t)**.
- Fourier Transforms of both signals were computed.
- The resulting spectra were convolved in the frequency domain.
- An Inverse Fourier Transform was applied to the convolution result.
- The obtained signal was compared with the direct time-domain multiplication result.

Results

- The frequency-domain convolution followed by IFT closely matched the direct time-domain multiplication.
- This confirmed the second form of the convolution theorem:

Conclusion

The experimental results obtained in LabVIEW showed strong agreement between the time-domain and frequency-domain operations. Therefore, both forms of the convolution theorem were successfully verified.



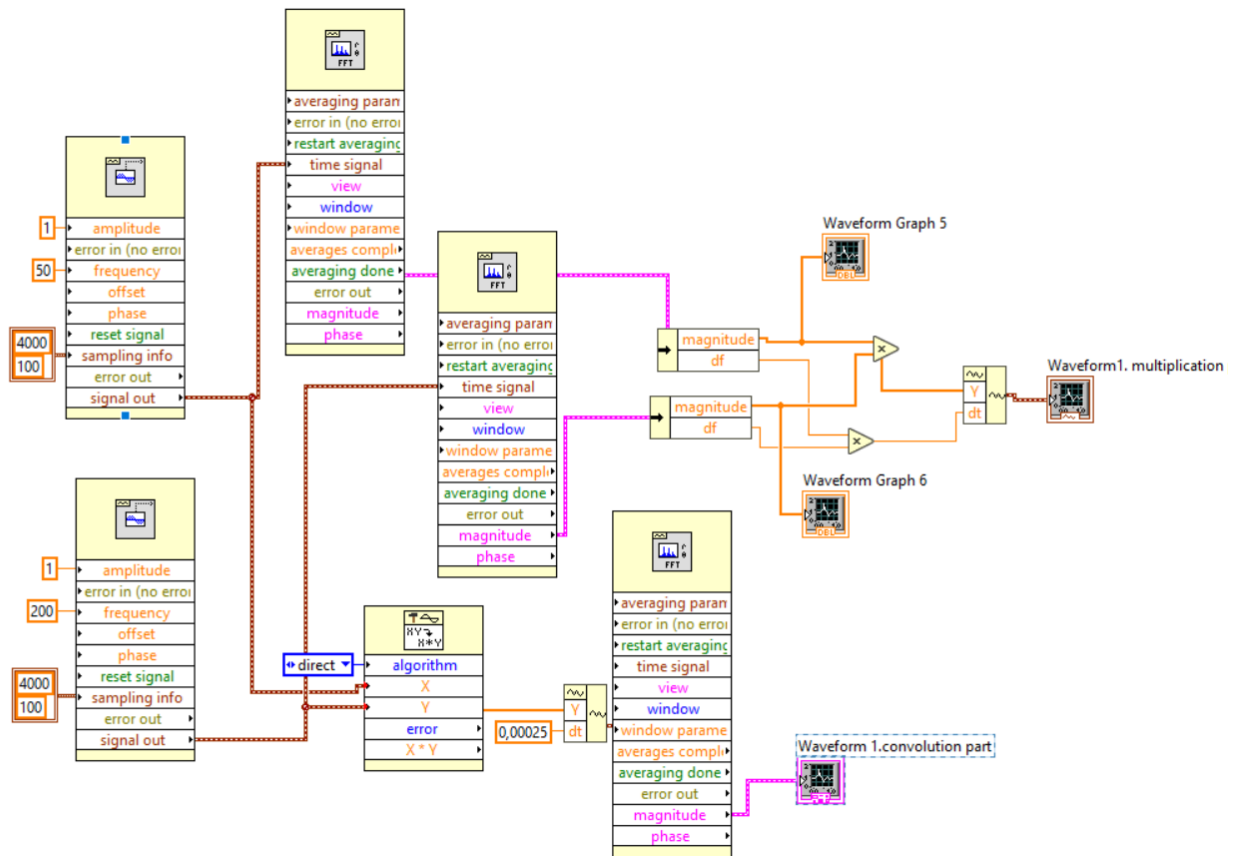
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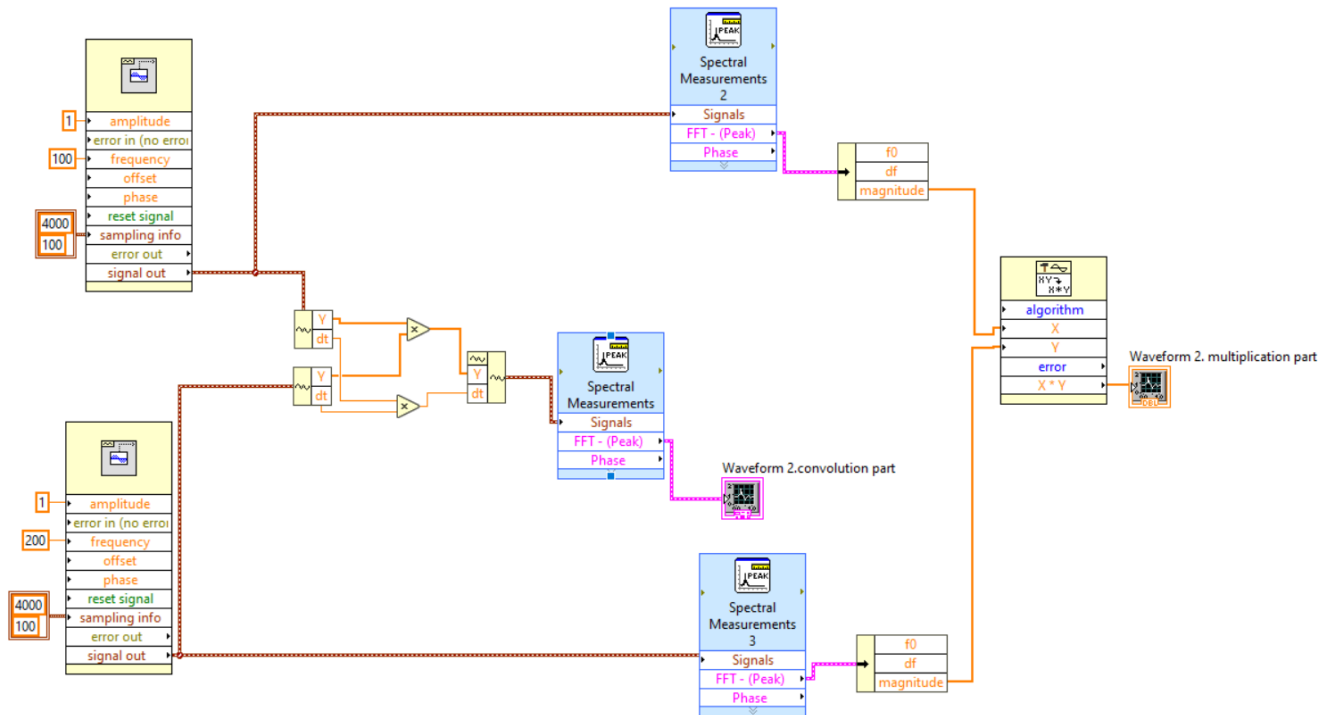
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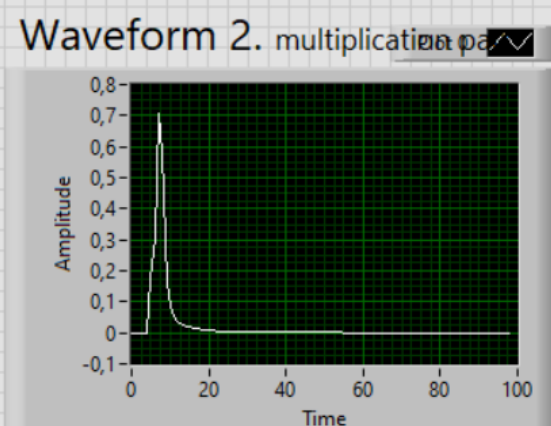
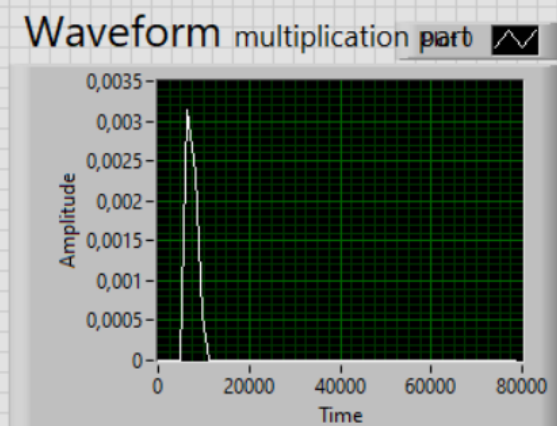
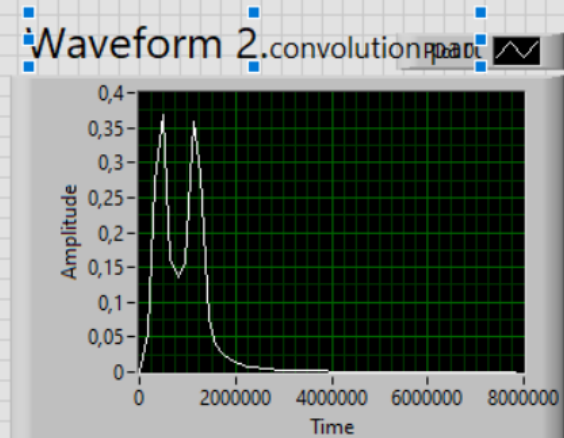
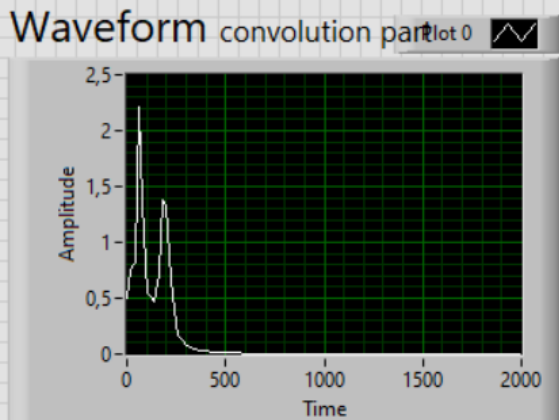
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Task 4: Alias Frequency Calculation

Objective

- To investigate the aliasing effect that occurs when a 2 kHz square wave is sampled at a frequency of 7 kHz and to identify the frequency components that appear in the FFT spectrum.

Methodology

- A square wave with a fundamental frequency of **2 kHz** was sampled at **7 kHz**.
- Since the Nyquist frequency is **3.5 kHz**, frequency components above this limit cannot be represented correctly and are reflected back into the observable frequency range.
- The harmonic components of the square wave were examined to determine their visible frequencies after sampling.

Results

- The fundamental component at **2 kHz** remained unchanged because it is below the Nyquist frequency.
- The **3rd harmonic (6 kHz)** appeared at approximately **1 kHz** due to aliasing.
- The **5th harmonic (10 kHz)** appeared at approximately **3 kHz**.
- The **7th harmonic (14 kHz)** folded back to **0 kHz**.
- As a result, the FFT spectrum was expected to contain significant peaks around **0 kHz, 1 kHz, 2 kHz, and 3 kHz**.

Conclusion

- The measured spectrum confirmed the presence of aliased harmonic components caused by sampling above the Nyquist limit.
- The observed frequency peaks corresponded to the folded versions of the original harmonics of the 2 kHz square wave.
- This experiment successfully demonstrated the aliasing phenomenon and its effect on frequency-domain analysis.



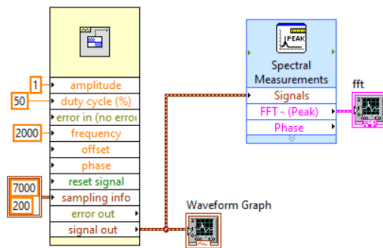
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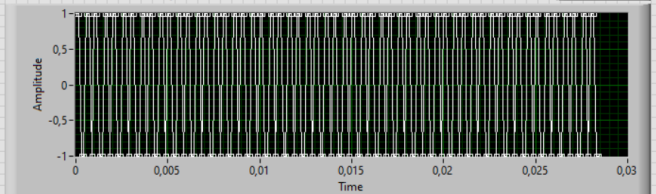
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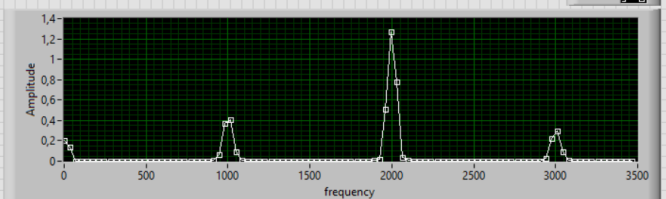
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Waveform Graph



fft





Exercise 2:

Using the LabVIEW software, create the following VIs with the NI-6008 board:

1. Perform the correct configuration of the board.
2. Generate and acquire a DC voltage.
3. Acquire an AC voltage at a frequency of 100 Hz generated by NI-6008.
4. Generate and acquire a digital signal using multiple digital outputs using NI-6008.
5. Using a waveform generator, generate a 1 kHz AC signal and perform spectral analysis using LabVIEW. Acquire the same signal with the digital oscilloscope and perform the FFT. Perform sampling synchronously with both acquisition instruments. Compare the obtained measurements. Describe the setup used to achieve synchronous sampling.
6. Generate a 1 kHz AC signal and perform spectral analysis. Sample asynchronously with both acquisition instruments. Apply one of the techniques designed to reduce scallop loss and leakage in both LabVIEW and the oscilloscope. Compare the obtained measurements.
7. Store in a text file, 1 s of signal acquired in AC at a frequency of 1 kHz. (use write to Measurement).
8. Read the file acquired above and display it on a waveform graph.



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Task 1: NI-6008 Board Configuration

Instruments

- NI LabVIEW Software
- NI-6008 Data Acquisition (DAQ) Board

Objective

- To configure the NI-6008 DAQ board and verify its proper operation before performing data acquisition and signal generation tasks.

Methodology

- The NI-6008 board was connected to the computer through the USB interface.
- The device was detected and selected using the NI driver software.
- A self-test procedure was performed to verify that the hardware was functioning correctly and communicating properly with the computer.
- The required acquisition parameters were configured according to the experimental setup.
- Both **Differential** and **RSE (Referenced Single-Ended)** input configurations were examined and adjusted as required.



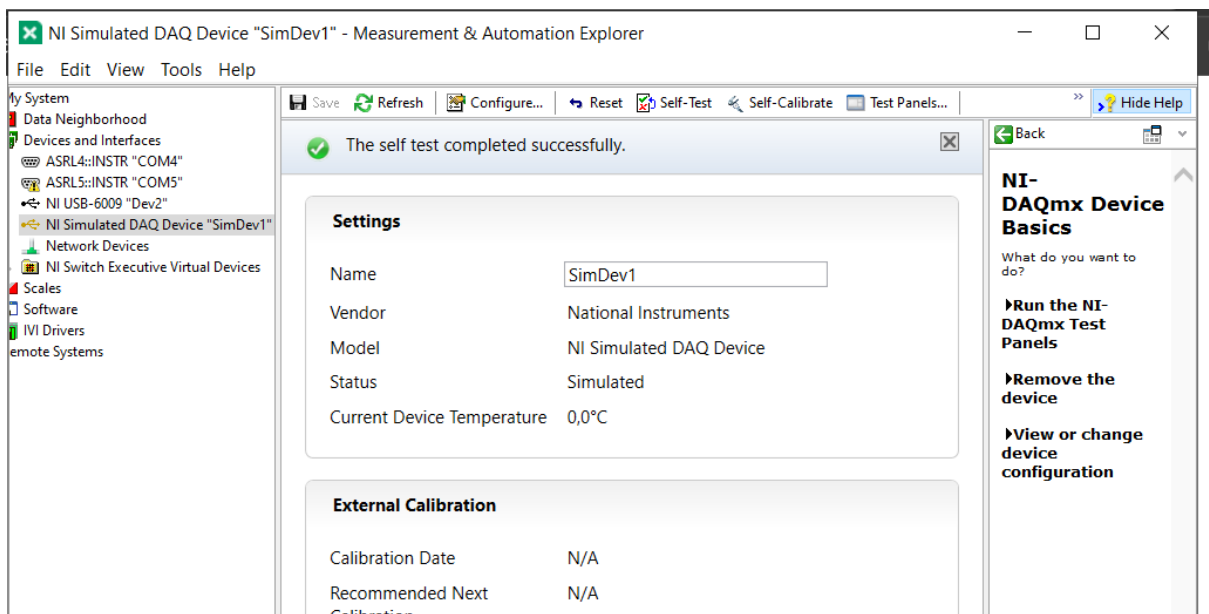
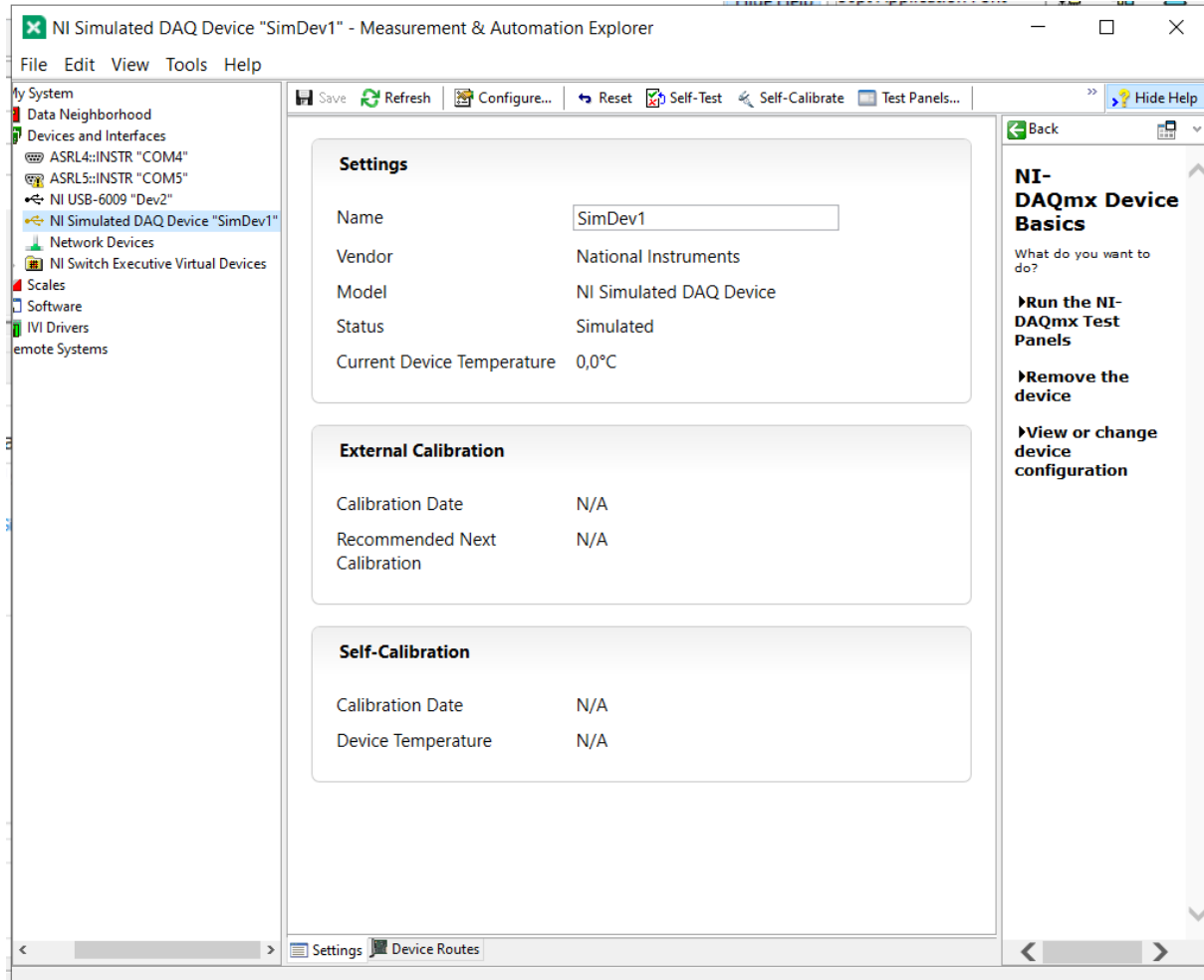
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Task 2: Designing a System for Generating and Acquiring DC Voltage Using DAQ

Objective

- To design and implement a DAQ-based system capable of generating, acquiring, and monitoring DC voltage signals in real time using LabVIEW.

Methodology

- The appropriate analog input and output channels of the DAQ device were selected for voltage generation and measurement.
- A variable DC voltage signal was generated through the analog output channel of the DAQ hardware.
- The generated voltage was connected to an analog input channel and continuously acquired by the DAQ system.
- A graphical interface was developed in LabVIEW to display the acquired voltage values.
- The generated and measured voltages were monitored in real time to verify the correct operation of the system.



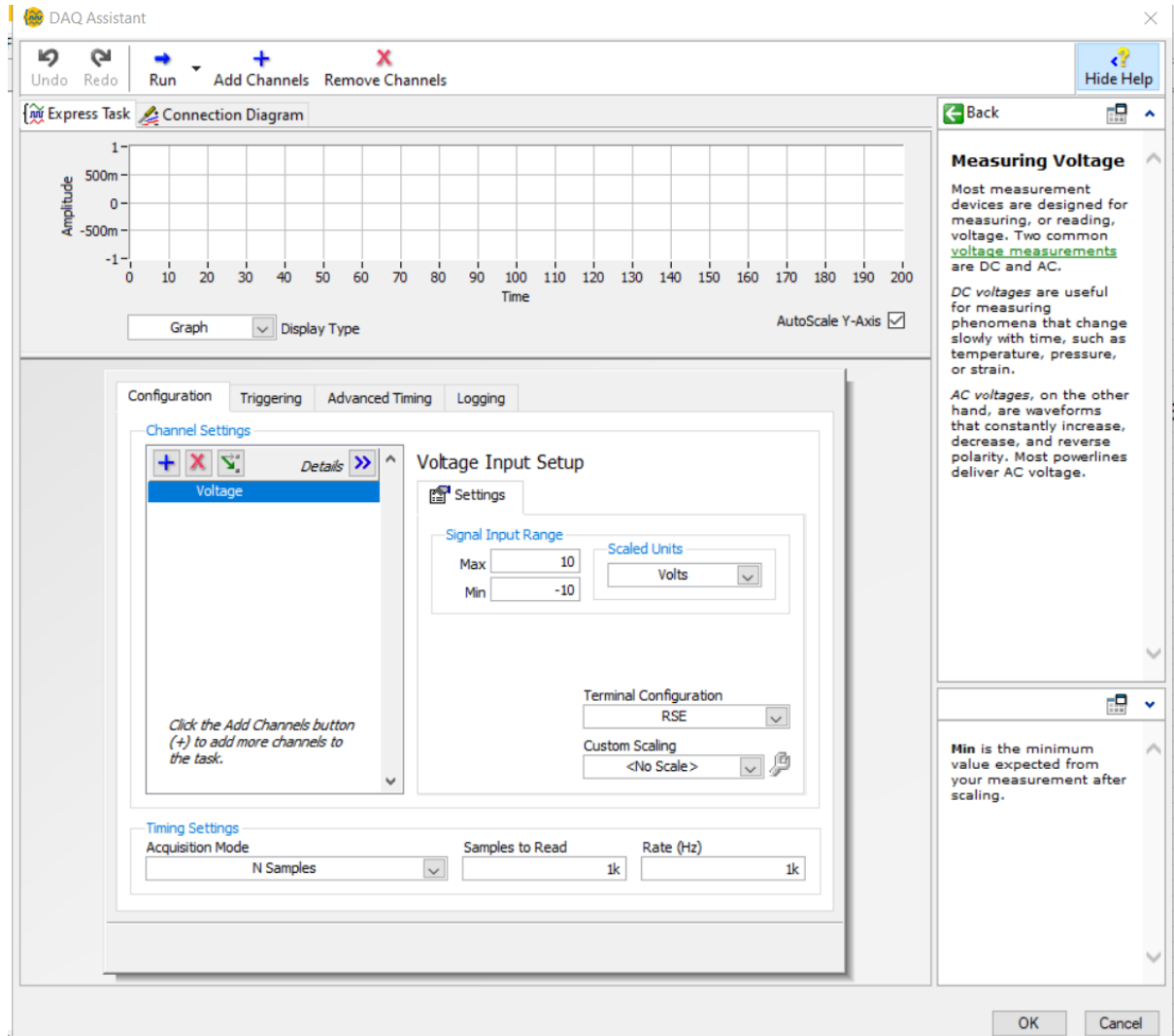
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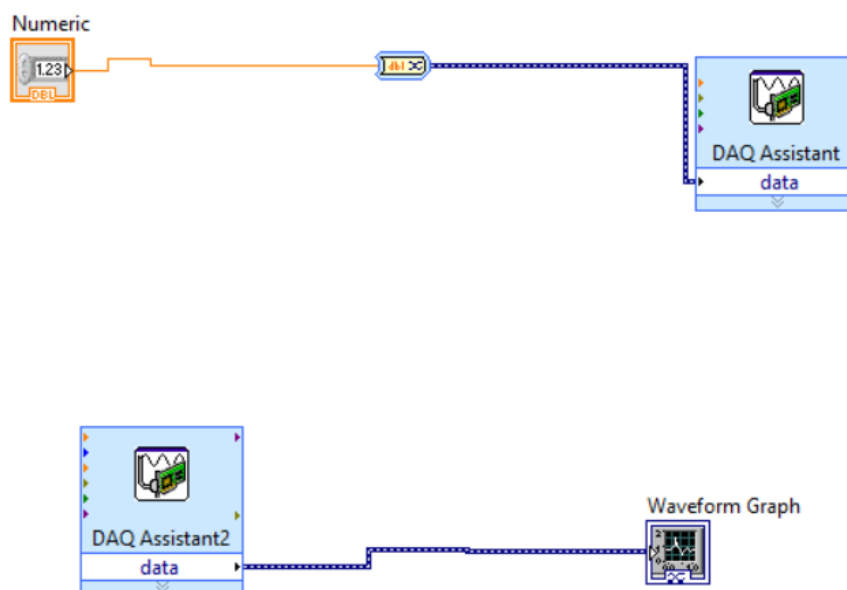
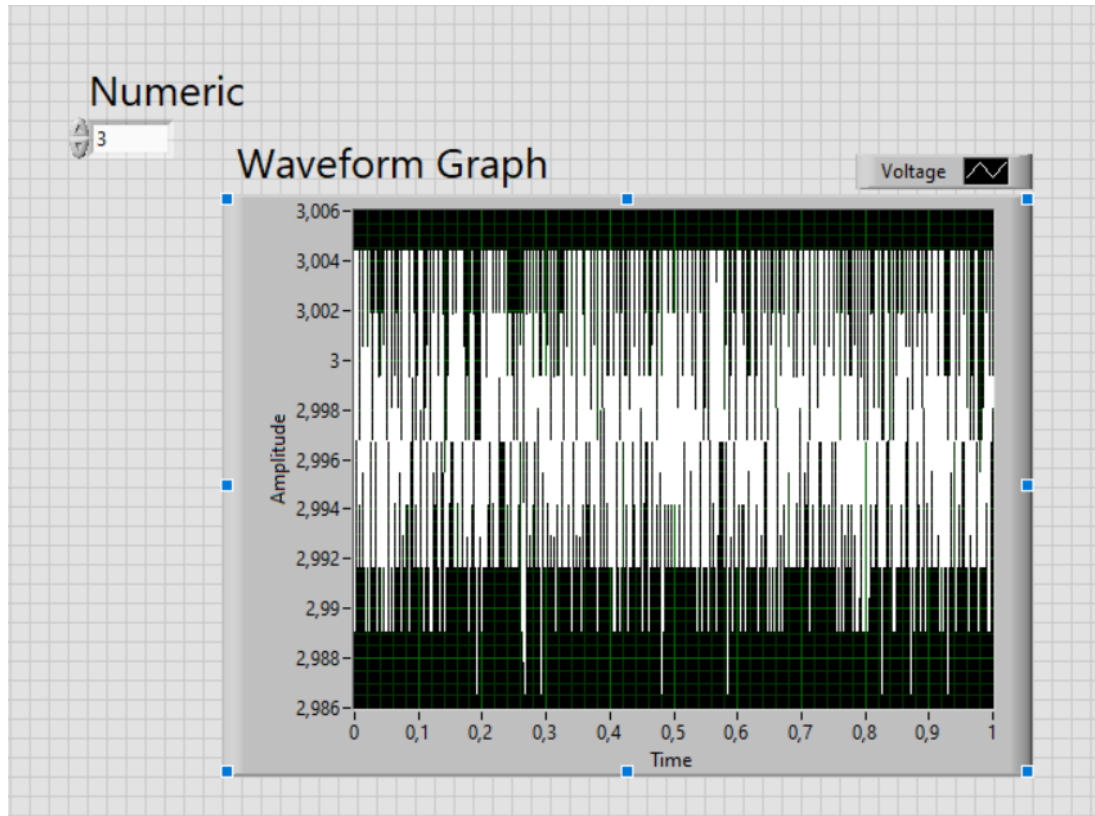
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Objective

- To generate a 100 Hz AC signal using LabVIEW and the NI-6008 DAQ device, acquire the generated signal, and analyze its characteristics in real time.

Methodology

- A 100 Hz sinusoidal AC signal was designed within the LabVIEW environment.
- The generated waveform was transmitted to the DAQ device through the analog output channel.
- The output signal was connected to an analog input channel and continuously acquired by the DAQ system.
- The acquired signal was displayed on a LabVIEW graph for real-time monitoring.
- The waveform characteristics, including amplitude, frequency, and sampled signal behavior, were observed and evaluated.



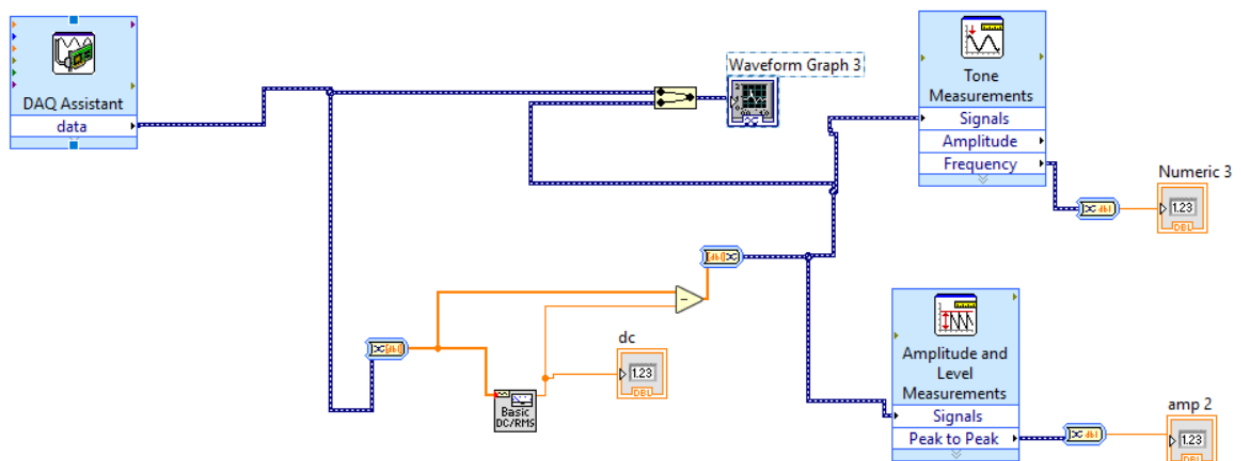
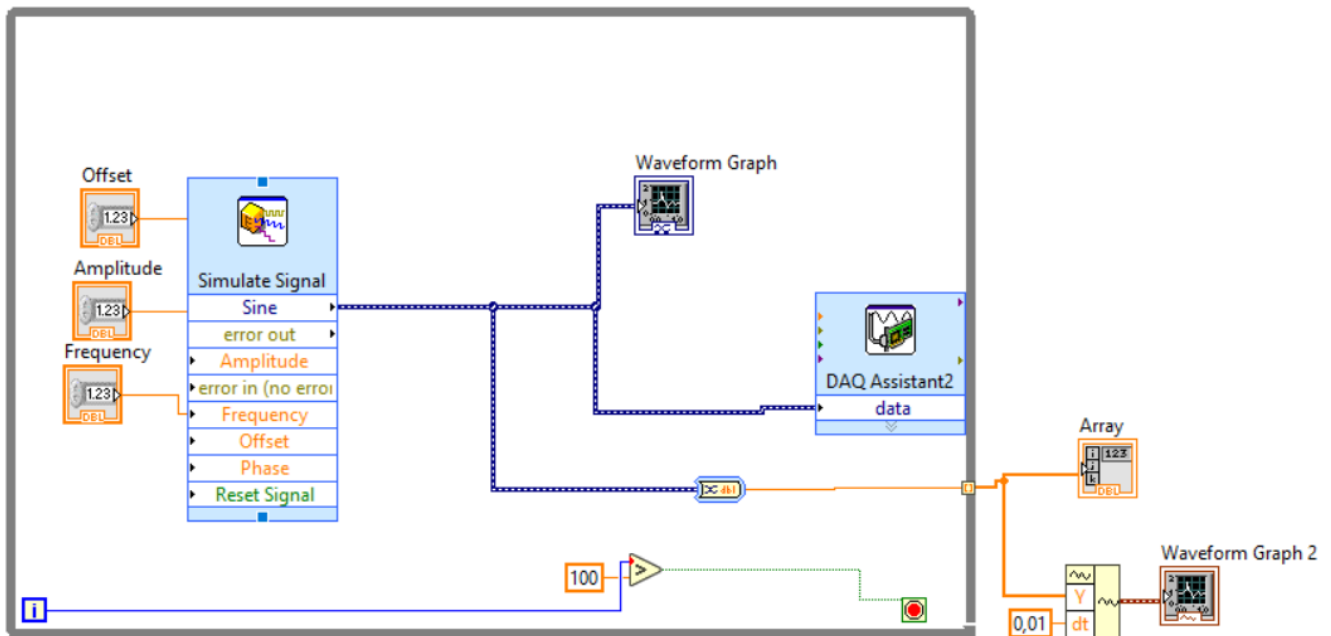
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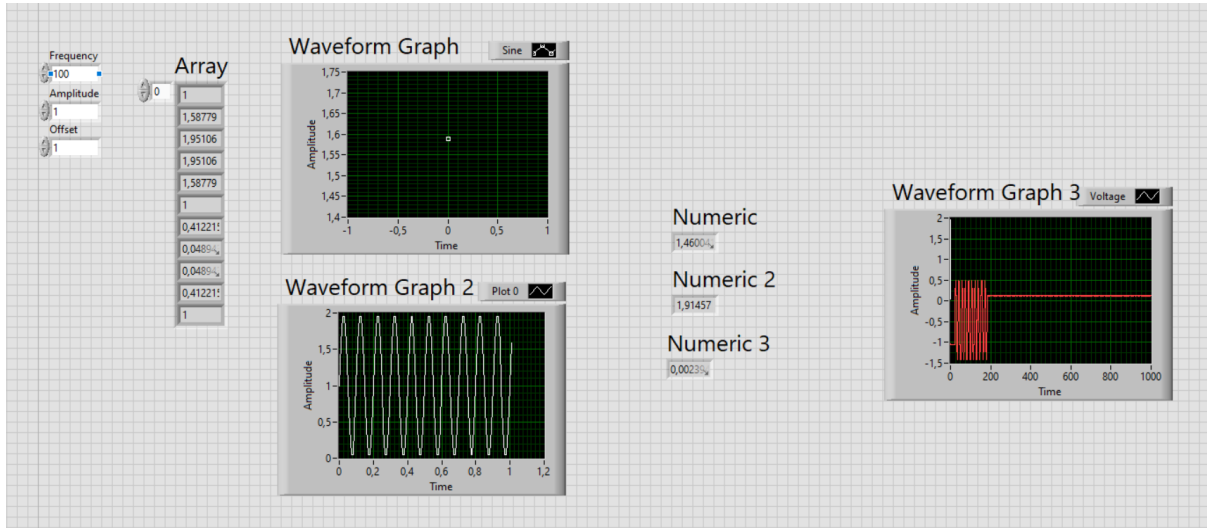
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Task 4: Generating and Acquiring a Digital Signal Using Multiple Digital Outputs with NI-6008

Objective

- To generate a digital signal using multiple digital output channels of the NI-6008 DAQ device, acquire the generated data through digital inputs, and verify the signal states using logic indicators in LabVIEW.

Methodology

- An array containing digital data patterns was created in LabVIEW.
- The array data was transmitted to the NI-6008 through multiple digital output channels.
- The generated digital signals were connected to the corresponding digital input channels of the DAQ device.
- The input channels continuously acquired the transmitted digital states.
- The acquired digital data was processed and displayed using logic indicators in LabVIEW.
- The transmitted and received digital patterns were compared to verify correct signal generation and acquisition.



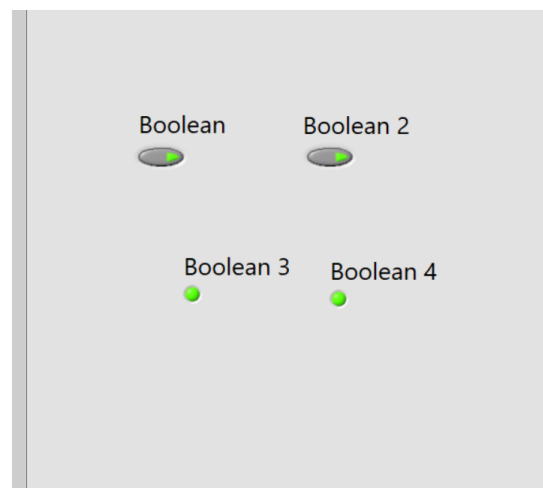
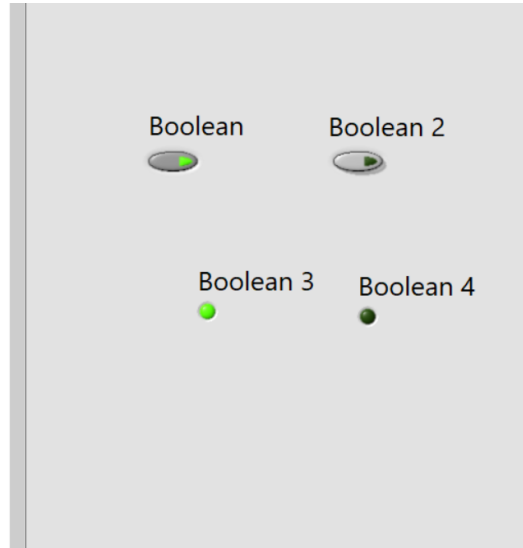
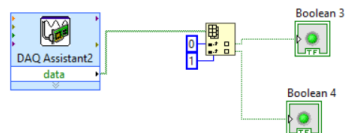
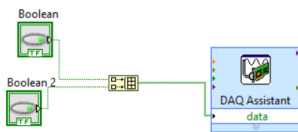
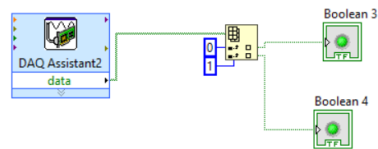
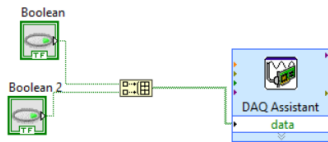
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Task 5 & 6: Spectral Analysis of a 1 kHz AC Signal Using LabVIEW and Digital Oscilloscope Instruments

- Tektronix AFG1022 Function Generator
- GW Instek GDS-1072A-U Digital Oscilloscope
- NI-6008 DAQ Device
- NI LabVIEW Software

Objective

- To generate a 1 kHz AC signal, perform spectral analysis using LabVIEW and a digital oscilloscope under both synchronous and asynchronous sampling conditions, and compare the obtained frequency-domain measurements.

Methodology

Task 5 – Synchronous Sampling

- A 1 kHz sinusoidal signal was generated using the Tektronix AFG1022 function generator.
- The generated signal was acquired through the NI-6008 DAQ device and processed in LabVIEW.
- FFT analysis was performed in LabVIEW to obtain the frequency spectrum of the signal.
- The same signal was simultaneously acquired using the GW Instek GDS-1072A-U digital oscilloscope.
- FFT analysis was performed using the oscilloscope's built-in spectral analysis function.
- Both acquisition systems were configured for synchronous sampling to ensure that the same signal was measured under identical conditions.
- Since the oscilloscope provided FFT amplitudes in dB, the comparison was carried out using dB values.

Task 6 – Asynchronous Sampling and Leakage Reduction

- The signal frequency was slightly shifted from the nominal 1 kHz value to create an asynchronous sampling condition.
- FFT measurements were repeated using both LabVIEW and the digital oscilloscope.
- To reduce spectral leakage and scallop loss, the window function was changed from **Rectangular** to **Hamming** in both instruments.
- The resulting spectra were analyzed and compared with the previous measurements.



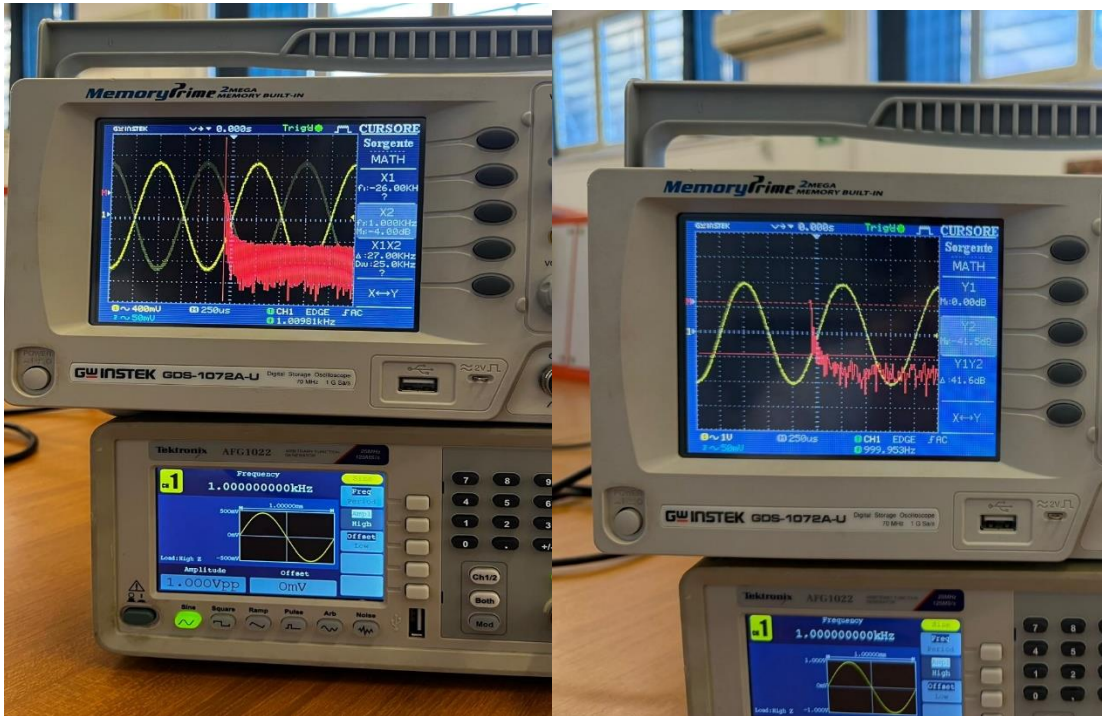
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Task 7 & 8: Signal Storage and Retrieval Using LabVIEW

Objective

- To acquire a 1-second AC signal, store the acquired data in a measurement file, and subsequently read the stored data for visualization and verification.

Methodology

Task 7 – Signal Storage

- A 1 kHz AC signal was acquired through the NI-6008 DAQ device.
- The acquired waveform was recorded for a duration of 1 second.
- The **Write to Measurement File** function in LabVIEW was used to save the acquired data into a text-based measurement file.
- The stored file contained the sampled waveform data for later analysis and processing.

Task 8 – Signal Retrieval and Visualization

- The measurement file generated in Task 7 was reopened using the corresponding file-reading component in LabVIEW.
- The stored waveform data was extracted from the file and reconstructed as a waveform signal.
- The retrieved signal was displayed on a waveform graph for visualization.
- The displayed waveform was compared with the original acquired signal to verify the integrity of the stored data.



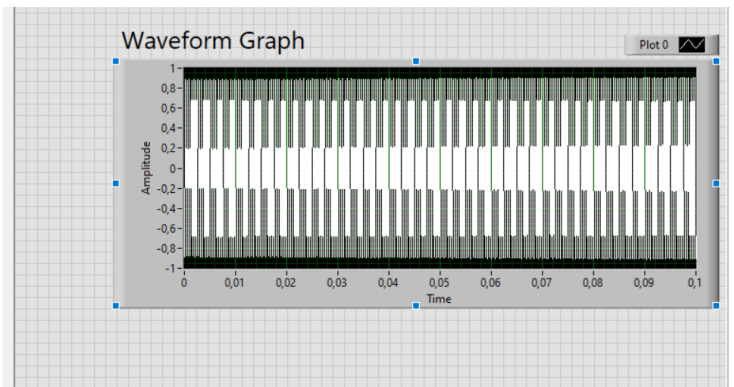
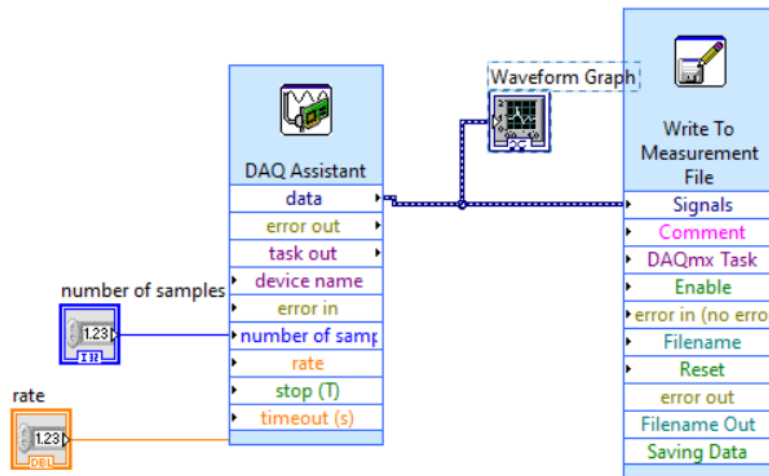
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Exercise 3:

Acquire via NI-6008 board a sinusoidal signal using a waveform generator:

1. Acquire 10 signal periods and measure in synchronous sampling: SNR, THD, SINAD, ENOB and SFDR of the sampler. Make measurements at the following generator frequencies: 50 Hz, 100 Hz, 500 Hz, 1 kHz, 2 kHz.
2. Repeat the measurement in asynchronous sampling and compare the results in step 3

Task 1&2: Performance Evaluation Under Synchronous Sampling

Objective

- To evaluate the performance of the acquisition system under synchronous sampling conditions by measuring key dynamic parameters at different input frequencies.

Methodology

- Sinusoidal signals with frequencies of **50 Hz, 100 Hz, 500 Hz, 1 kHz, and 2 kHz** were generated.
- For each frequency, **10 signal periods** were acquired under synchronous sampling conditions.
- FFT-based spectral analysis was performed on the acquired data.
- The performance parameters **SNR (Signal-to-Noise Ratio)**, **THD (Total Harmonic Distortion)**, **SINAD (Signal-to-Noise and Distortion Ratio)**, **ENOB (Effective Number of Bits)**, and **SFDR (Spurious-Free Dynamic Range)** were calculated according to their respective formulations.

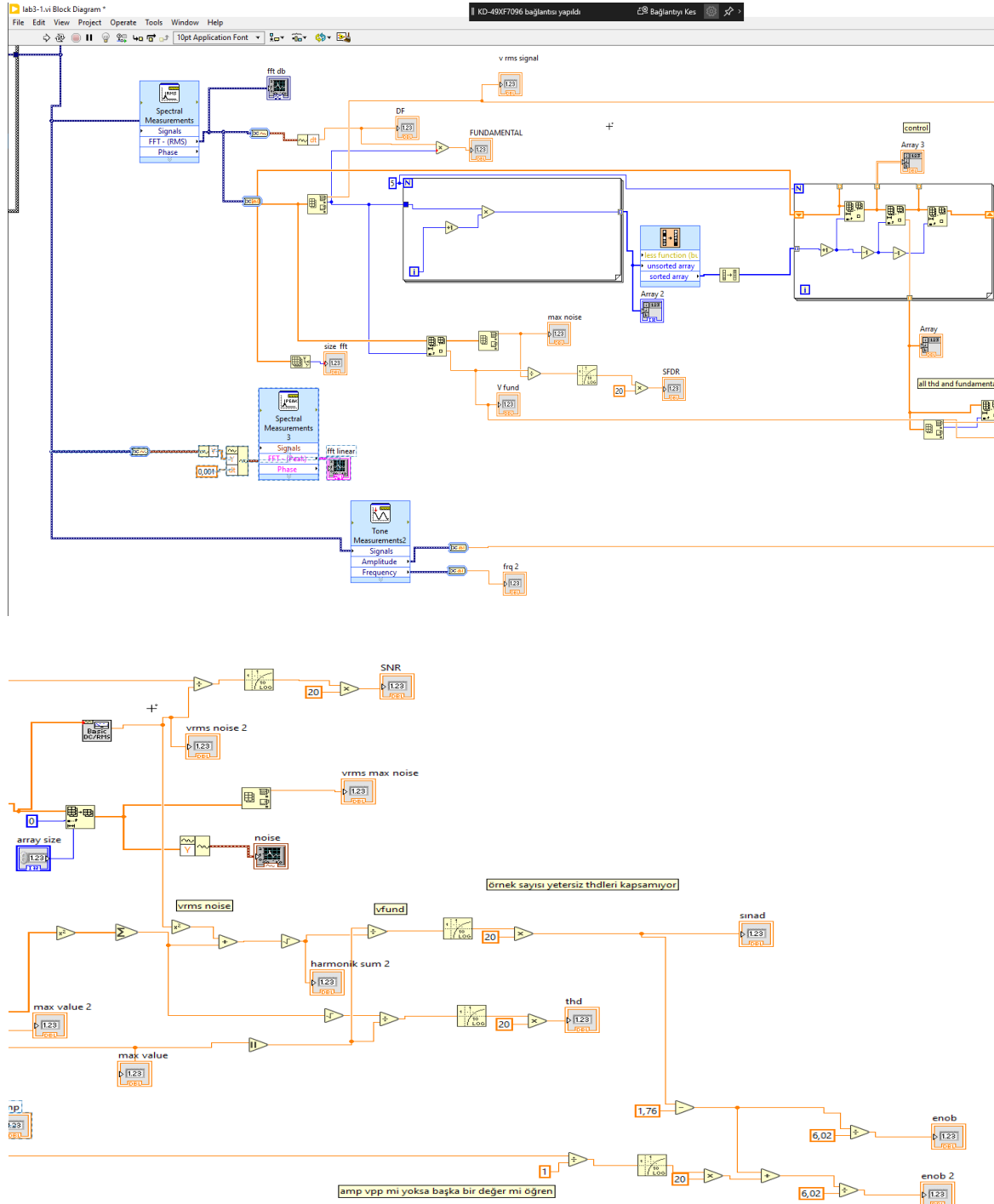


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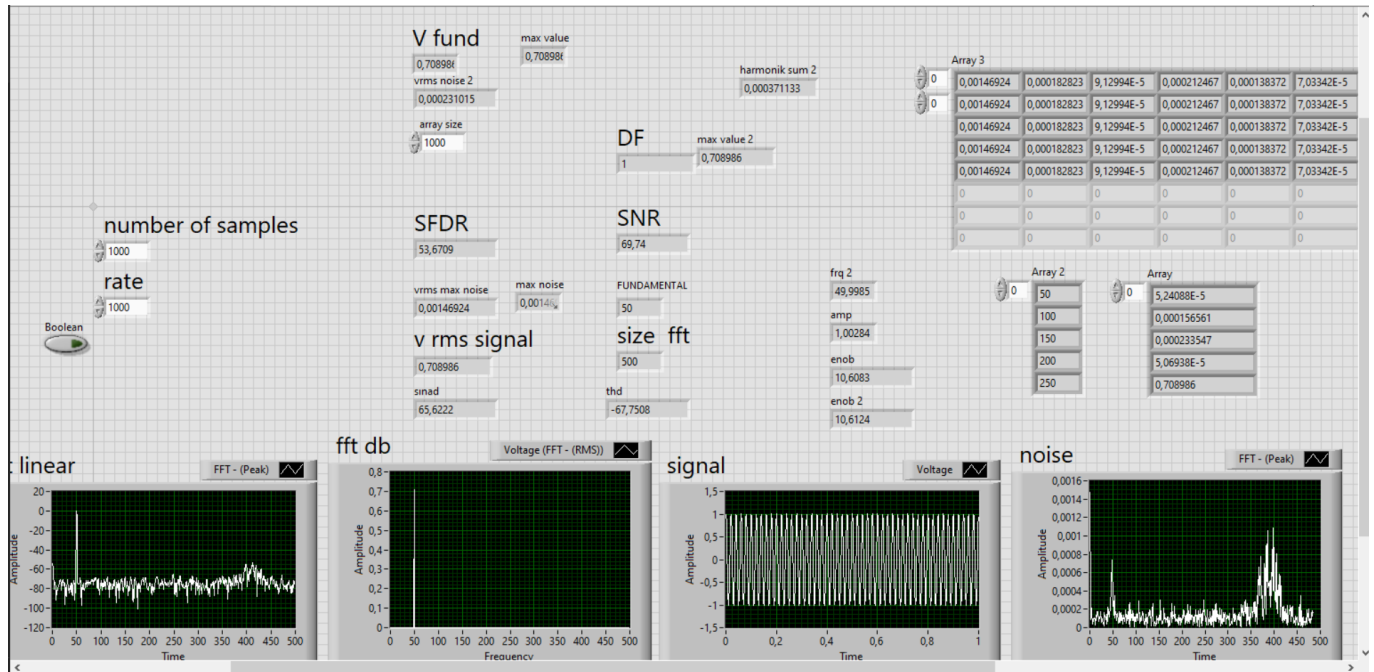
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50hz synchron





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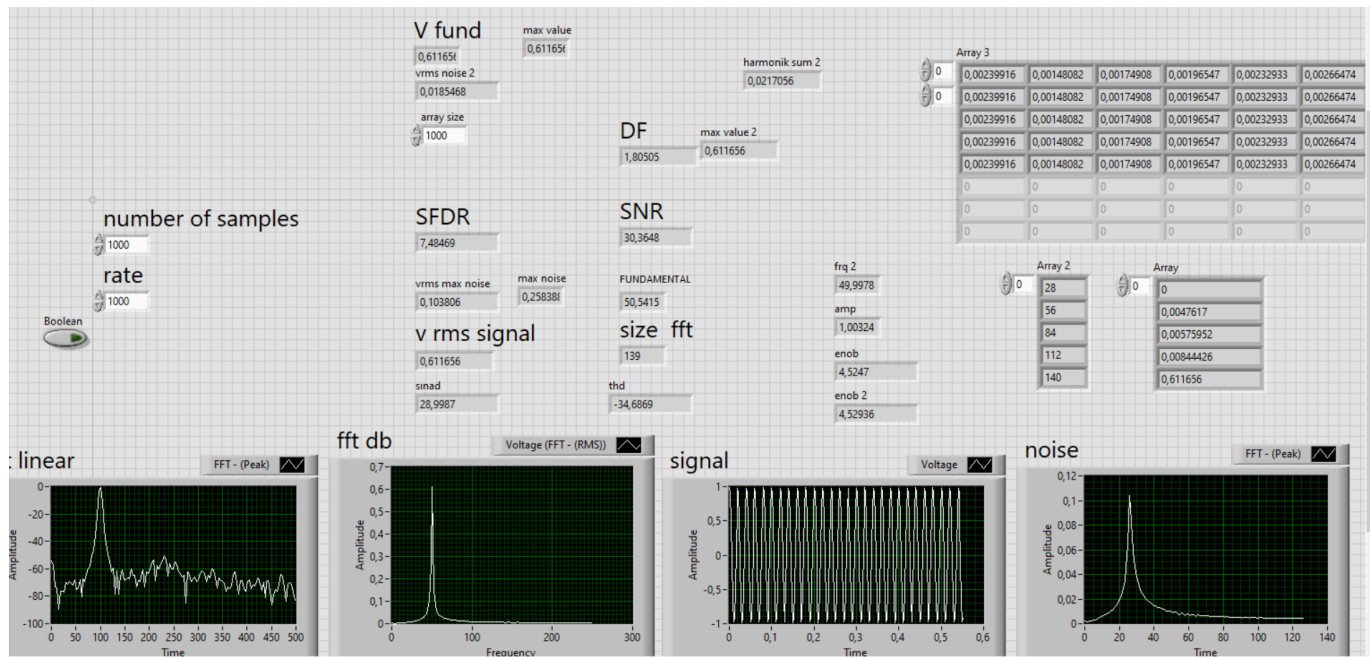
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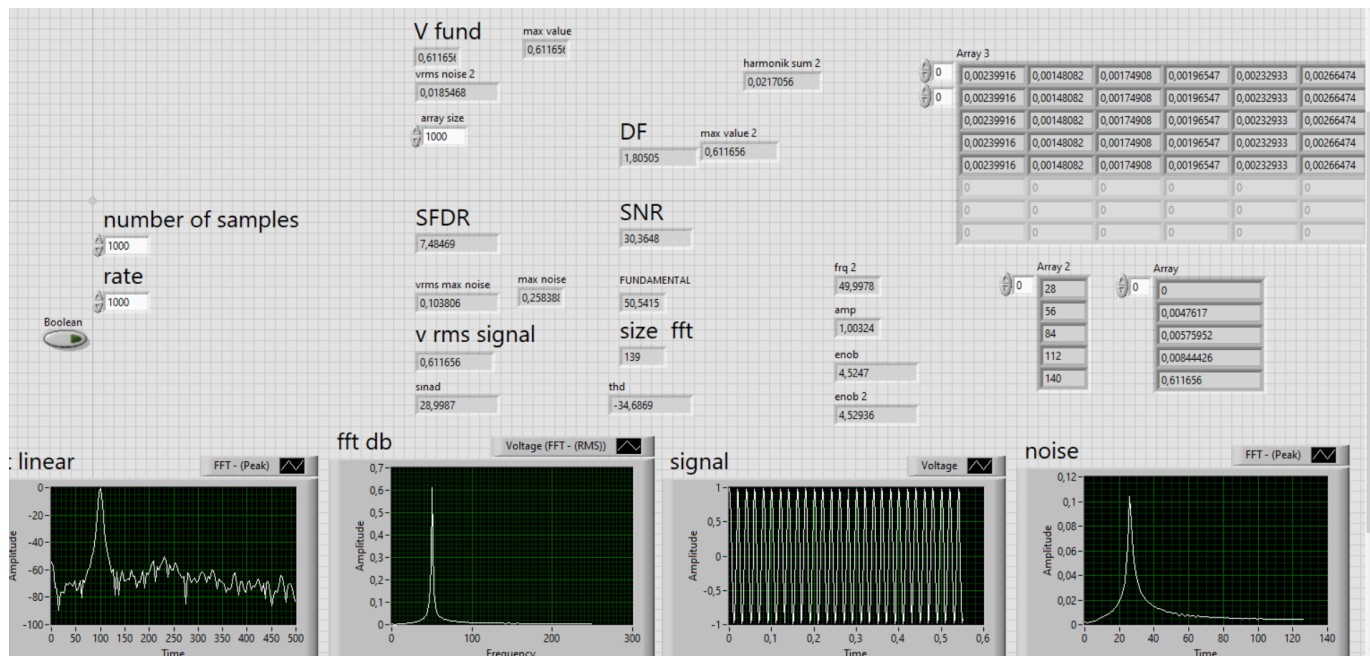
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50hz asynchronous



100 hz sync





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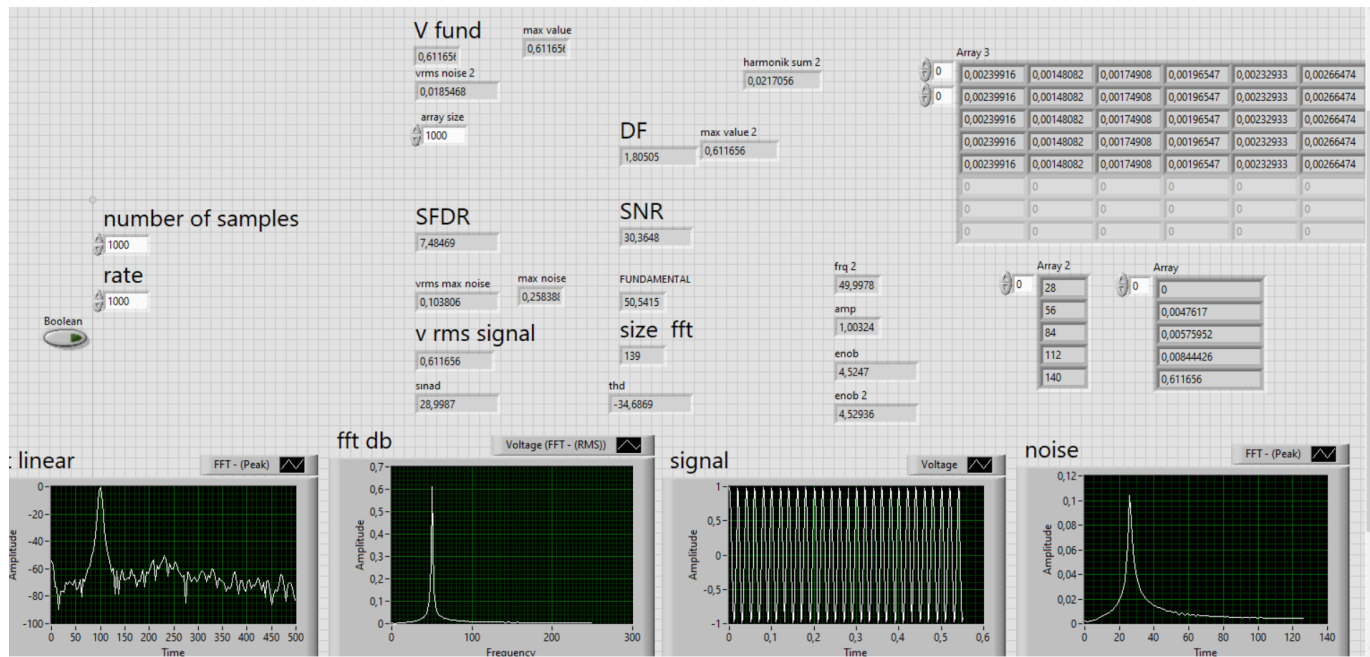
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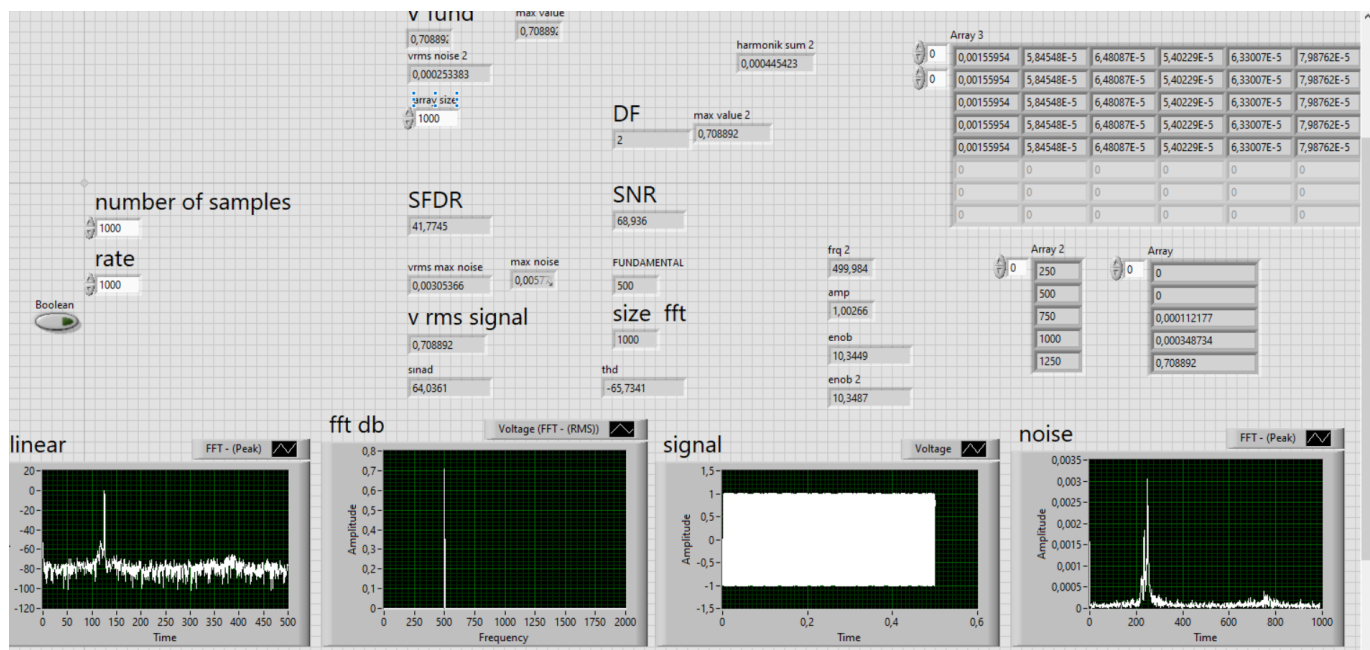
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100 hz async



500 hz sync





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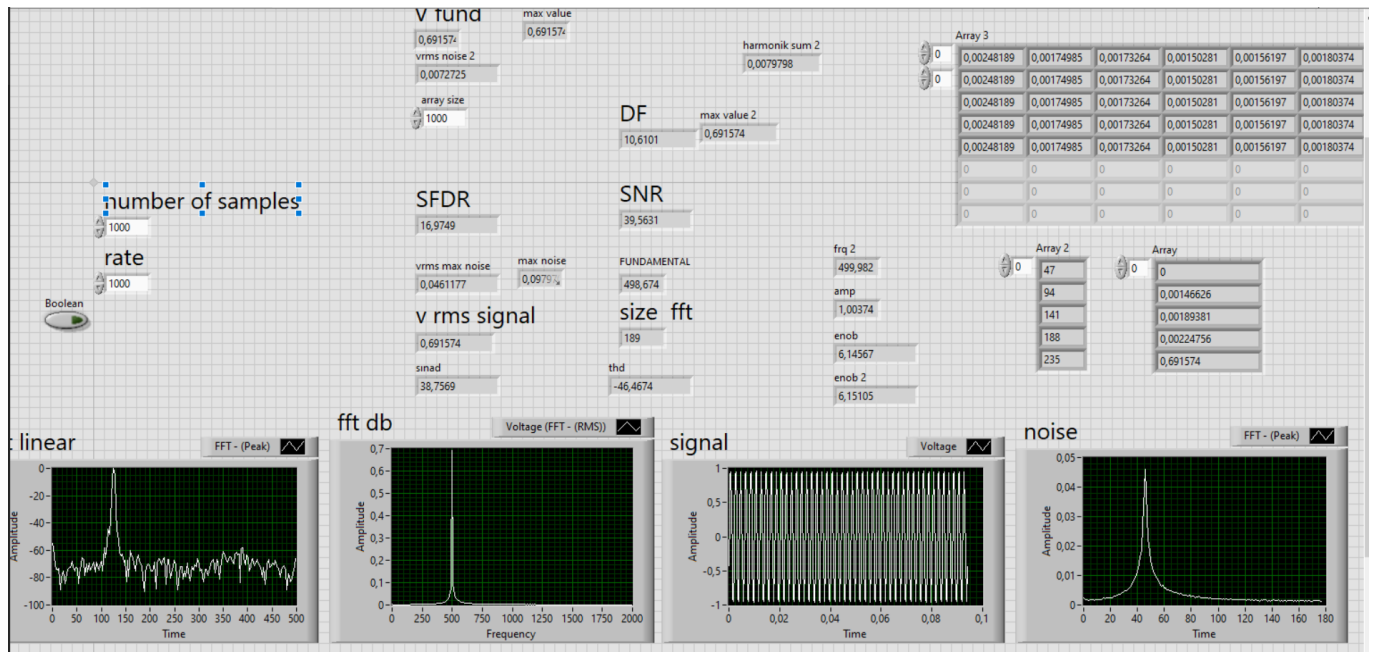
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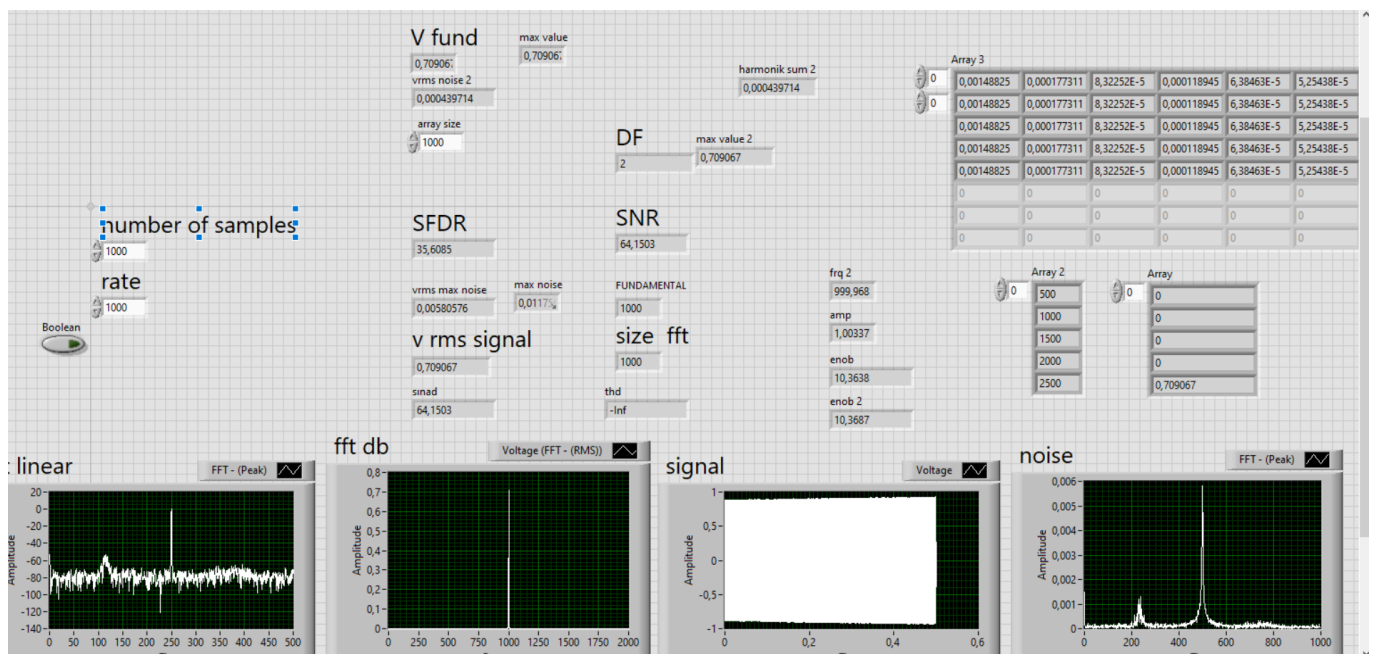
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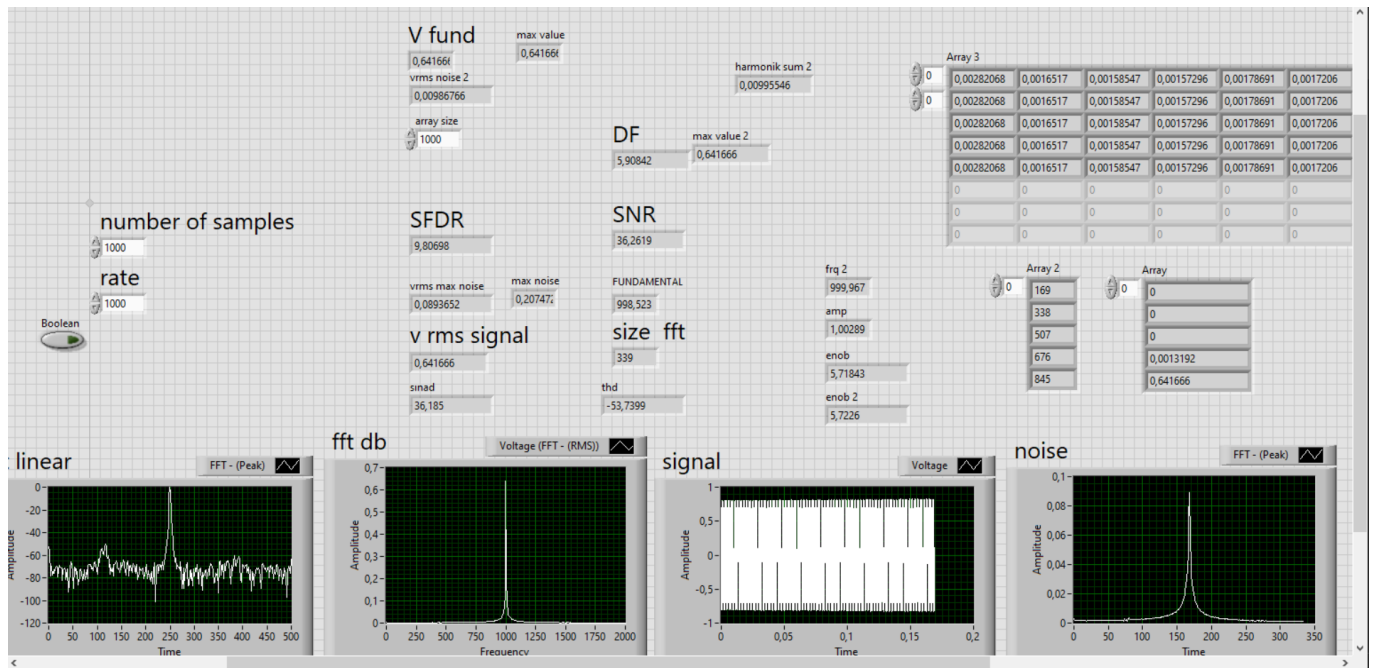
500 hz async



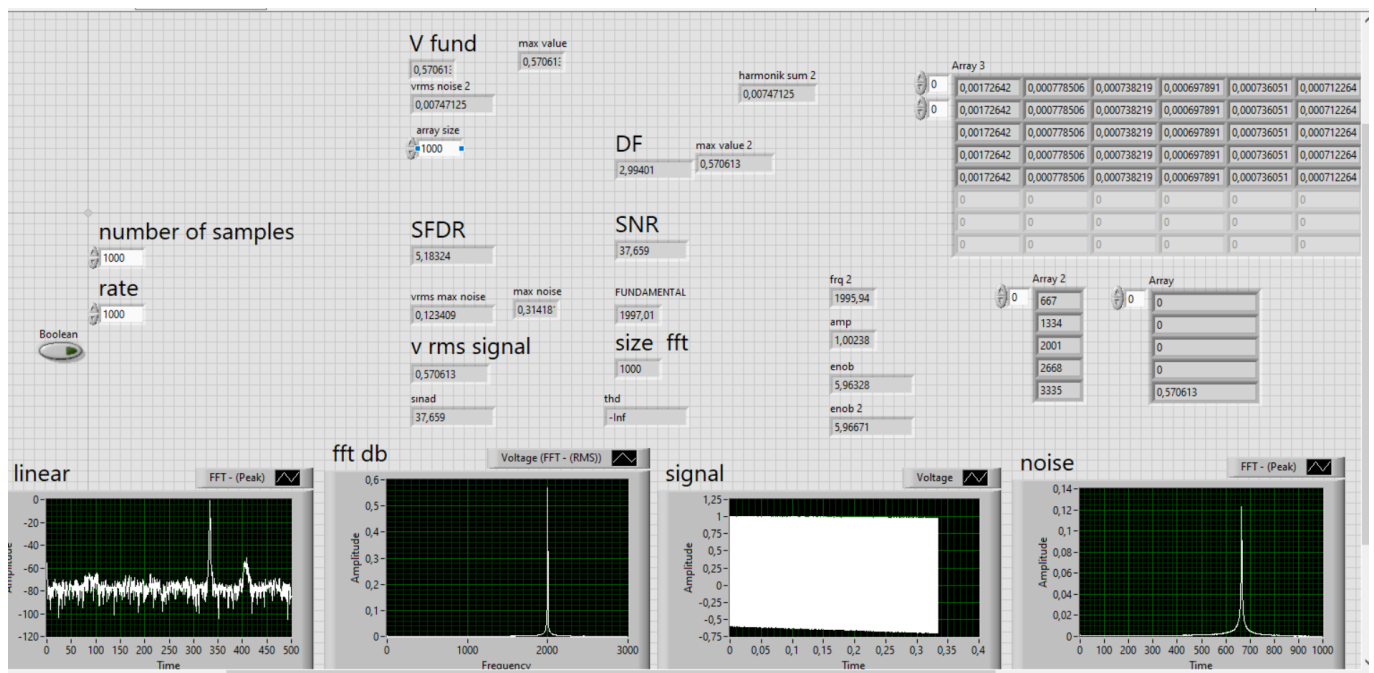
1khz sync



1khz async



2khz sync





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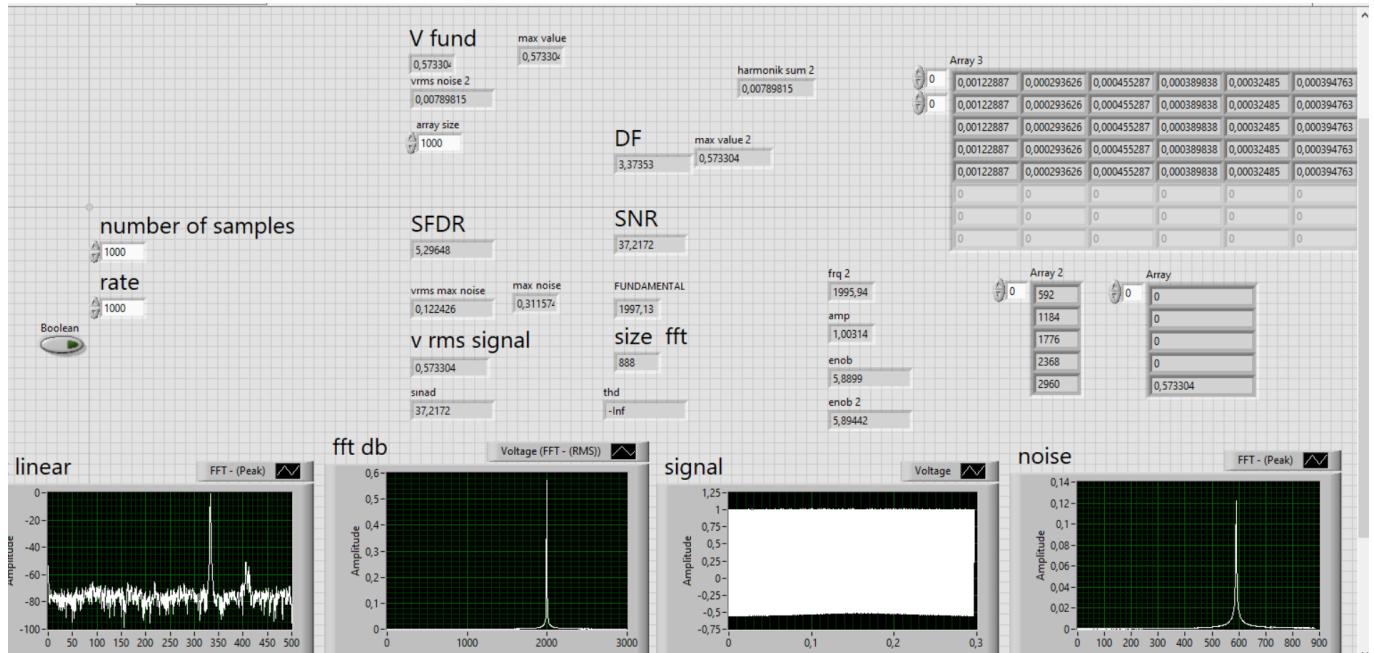
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2khz async



Exercise 4:

Carry out the measurement of small resistance with the volt-amperometric method. Use two digital multimeters driven by LabVIEW. Set the multimeters correctly to measure voltage and current. Set the full-scale values correctly. Make:

1. The category A uncertainty evaluation of the resistance by carrying out 100 voltage and current measurements simultaneously. Correctly indicate the number of significant digits.
2. The category B uncertainty evaluation by choosing one measurement from the 100 previously carried out. Correctly indicate the number of significant digits.



Task 1 & 2 Measurement of Small Resistance Using the Volt-Amperometric Method

Instruments

- GPS-8303 Power Supply
- 2 × SDM3045X Digital Multimeters
- NI LabVIEW Software

Objective

- To measure a small resistance using the volt-amperometric method and evaluate both Category A and Category B uncertainties of the measurement.

Methodology

- A measurement circuit was assembled using the GPS-8303 power supply and the resistance under test.
- Two SDM3045X digital multimeters were configured to simultaneously measure voltage and current.
- Appropriate measurement ranges were selected to ensure accurate readings.
- A total of **100 simultaneous voltage and current measurements** were acquired through LabVIEW.
- Resistance values were calculated from the measured voltage and current data.
- The **Category A uncertainty evaluation** was performed in LabVIEW using statistical analysis of the 100 acquired measurements.
- For the **Category B uncertainty evaluation**, one measurement set was selected from the acquired data.
- The uncertainty calculation was carried out using the accuracy specifications and resolution values provided in the SDM3045X datasheet.
- All results were reported with the appropriate uncertainty values and the correct number of significant digits.



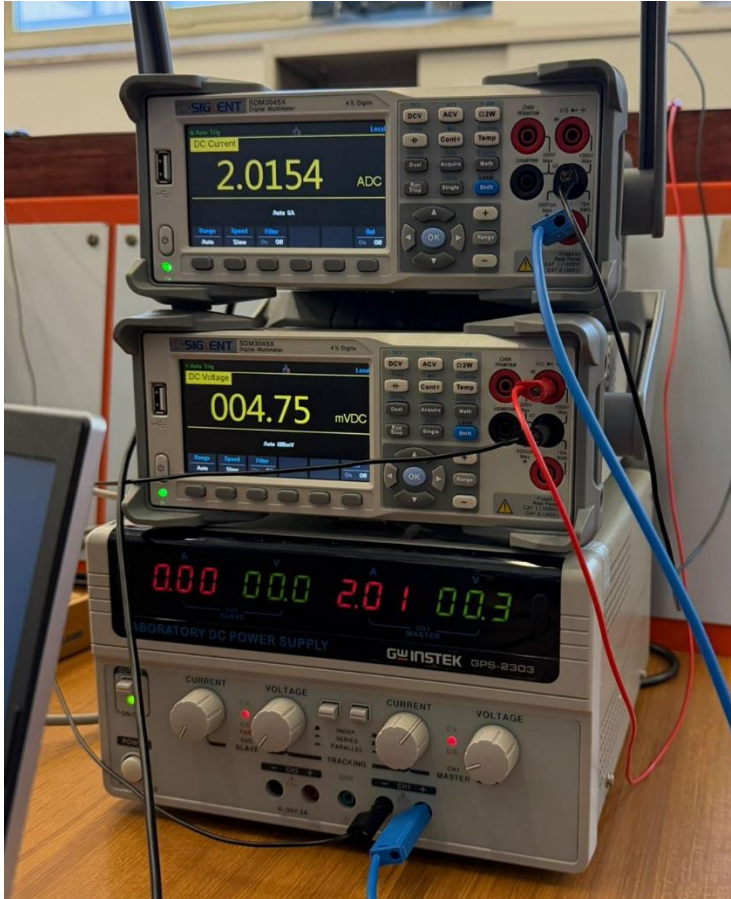
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sum	Best estimate of the mea	Array	Experimental waste
0.453538	0.00453538	0.0045	0.0000167721
		0.0045	Absolute uncertainty
		0.0045	0.00000167721
		0.0045	Relative uncertainty value (%)
		0.0045	0.0369806
		0.0045	
		0.0045	



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Dmm Datasheet

Function	Range (*2)	Resolution	Input Resistance etc.	Accuracy(*1) 1 Year 23°C±5°C
DC Voltage	500.00 mV	10µV	10MΩ or >10GΩ	0.02 + 4
	5.0000 V	100µV	10MΩ or >10GΩ	0.02 + 4
	50.000 V	1mV	11.11MΩ	0.02 + 4
	500.00 V	10mV	10.1MΩ	0.02 + 4
	1000.0 V	100mV	10MΩ	0.02 + 4
Resistance	500.00 Ω	10mΩ	0.83mA	0.10 + 5 (*3)
	5.0000 kΩ	100mΩ	0.83mA	0.10 + 3 (*3)
	50.000 kΩ	1Ω	83µA	0.10 + 3
	500.00 kΩ	10Ω	8.3µA	0.10 + 3
	5.0000 MΩ	100Ω	830nA	0.10 + 3
	50.000 MΩ	1kΩ	560nA//10 MΩ	0.30 + 3
DC Current	500.00 µA	10nA	0.06V max.	0.05 + 5
	5.0000 mA	100nA	0.6V max.	0.05 + 4
	50.000 mA	1µA	0.14V max.	0.05 + 4
	500.00 mA	10µA	1.4V max.	0.10 + 4
	5.0000 A	100µA	0.5V max.	0.25 + 5
	10.000 A	1mA	0.8V max.	0.25 + 5
Continuity	5000.0 Ω	100mΩ	0.83mA	0.10 + 5
Diode Test	5.0000 V	100µV	0.83mA	0.05 + 5
Capacitance	5.000 nF: 0.5~1nF	0.001nF	8.3µA	2.00 + 20
	5.000 nF: 1~5nF	0.001nF	8.3µA	2.00 + 10
	50.00 nF: 5~10nF	0.01nF	8.3µA	2.00 + 30
	50.00 nF: 10~50nF	0.01nF	8.3µA	2.00 + 10
	500.0 nF	0.1nF	83µA	2.00 + 4
	5.000 µF	1nF	0.56mA	2.00 + 4
	50.00 µF	10nF	0.83mA	2.00 + 4
True RMS Voltage AC (or AC+DC) (*4/*5/*6)	500.00 mV	10µV	30Hz – 50Hz	1.00 + 40
			50Hz – 10kHz	0.50 + 40
			10kHz – 30kHz	2.00 + 60
			30kHz – 100kHz	3.00 + 120
	5.0000 V	100µV	30Hz – 50Hz	1.00 + 20
			50Hz – 10kHz	0.35 + 15



We found: the Current = 2.0185 A The accuracy => 0.25 + 5 The resolution => 100 μ A

Digit uncertainty:

Relative uncertainty:

$$\mu_{I1} = \frac{5 \cdot 100 \cdot 10^{-6}}{2.0185 \cdot \sqrt{3}} = 1.43 \cdot 10^{-4} \text{ p.u} = 0.0143 \%$$

Absolute uncertainty:

$$\mu_{I1} = \frac{5 \cdot 100 \cdot 10^{-6}}{\sqrt{3}} = 2.89 \cdot 10^{-4} \text{ A}$$

Read uncertainty:

Relative uncertainty:

$$\mu_{I2} = \frac{0.25}{100 \cdot \sqrt{3}} = 0.0014 \text{ p.u} = 0.14 \%$$

Absolute uncertainty:

$$\mu_{I2} = \frac{0.25 \cdot 2.0185}{100 \cdot \sqrt{3}} = 0.0029 \text{ A}$$

Composed uncertainty:

$$\mu_i = \sqrt{(\mu_{I1}^2 + \mu_{I2}^2)} = \sqrt{(0.0143^2 + 0.14^2)} = 0.141 \%$$

$$\mu_I = \sqrt{(\mu_{I1}^2 + \mu_{I2}^2)} = \sqrt{((2.89 \cdot 10^{-4})^2 + (0.0029)^2)} = 2.914 \text{ mA}$$



We found: the voltage = 9.15 mV The accuracy => 0.02 + 4 The resolution => 10 μ V

Digit uncertainty:

Relative uncertainty:

$$\mu_{v1} = \frac{4 \cdot 10 \cdot 10^{-6}}{9.15 \cdot 10^{-3} \cdot \sqrt{3}} = 0.25 \%$$

Absolute uncertainty:

$$\mu_v = \left(\frac{4 \cdot 10 \cdot 10^{-6}}{\sqrt{3}} \right) = 2.31 \cdot 10^{-5} V$$

Read uncertainty:

Relative uncertainty:

$$\mu_v = \left(\frac{0.02}{100 \cdot \sqrt{3}} \right) = 0.00012 \text{ p.u.} = 0.012 \%$$

Absolute uncertainty:

$$\mu_{v2} = \frac{0.02 \cdot 9.15 \cdot 10^{-3}}{100 \cdot \sqrt{3}} = 1.056 \cdot 10^{-6} V$$

Composed uncertainty:

$$\mu_v = \sqrt{(\mu_{v1}^2 + \mu_{v2}^2)} = \sqrt{(0.25^2 + 0.012^2)} = 0.25 \%$$

$$\mu_v = \sqrt{(\mu_{v1}^2 + \mu_{v2}^2)} = \sqrt{((2.31 \cdot 10^{-5})^2 + (1.056 \cdot 10^{-6})^2)} = 2.3124 \cdot 10^{-5} V$$



Exercise 5:

Using an LCR Meter, perform A and B category uncertainty assessment of passive devices as indicated below. For A category uncertainty, carry out at least 100 measurements.

1. Measure a resistance at 50Hz, 10kHz, 300kHz
2. Measure non-polarized capacitance at 50Hz, 10kHz and 300kHz
3. Measure electrolyte capacitance at 50Hz, 10kHz and 300kHz
4. Measure 22mH inductance at 50Hz, 10kHz, 300kHz. Also measure the L_s - R_s at $f=50$ Hz.

Task 1: Uncertainty Assessment of Passive Components Using an LCR Meter

Instruments

- LCR-6300 LCR Meter
- NI LabVIEW Software

Objective

- To measure the electrical parameters of passive components and evaluate both Category A and Category B uncertainties of the measurements.

Methodology

- Measurements were performed using the LCR-6300 LCR meter at **50 Hz, 10 kHz, and 300 kHz**.
- The following passive components were characterized:
 - Resistance
 - Non-polarized capacitor
 - Electrolytic capacitor
 - 22 mH inductor
- For the inductor, the **L_s - R_s** parameters were additionally measured at **50 Hz**.
- For each measurement condition, at least **100 samples** were acquired.
- The measured values were transferred to LabVIEW and processed for statistical analysis.
- **Category A uncertainty** was calculated in LabVIEW using the acquired measurement dataset.
- **Category B uncertainty** was evaluated using the accuracy specifications and measurement limits provided by the LCR meter datasheet.
- All reported values were expressed using the appropriate uncertainty and correct number of significant digits.



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Resistance Measurement Results

The resistance measurement began by identifying the nominal value of the resistor through its color code in order to estimate the expected reading. A resistor with a nominal value of $50\ \Omega$ was selected and connected to the LCR meter using the appropriate test clips. Measurements were performed at 50 Hz, 10 kHz, and 300 kHz. At 50 Hz, the measured resistance was approximately $47\ \Omega$, while the reactance was found to be very small and capacitive in nature. As the measurement frequency increased to 10 kHz and 300 kHz, the reactance became purely inductive. This behavior can be attributed to the parasitic inductance associated with the resistor and its leads, which becomes more significant at higher frequencies.

Electrolytic Capacitor Measurement Results

The electrolytic capacitor was characterized at different frequencies to evaluate its frequency-dependent behavior. At low and intermediate frequencies, the component exhibited the expected capacitive characteristics. However, at 300 kHz, a negative capacitance value was observed. This result indicates that the operating frequency exceeded the capacitor's Self-Resonant Frequency (SRF), causing the parasitic inductive effects to dominate its electrical behavior. Consequently, the component no longer behaved as a capacitor but rather as an inductor. For this reason, the equivalent circuit model was changed from a capacitive model to an inductive model when analyzing the measurement results at 300 kHz.



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The screenshot displays a software interface for electronic measurements, featuring a grid background. The interface is organized into several sections:

- VISA resource name:** A dropdown menu showing "COM5".
- Serial Settings: Baud Rate:** A numeric input field set to "115200".
- Function (0: Cs-Rs):** A dropdown menu showing "Cs-Rs" with a value of "0".
- Frequency (1000):** A numeric input field set to "1000".
- Level Mode:** A dropdown menu showing "Voltage" with a value of "0".
- Voltage Level (1):** A numeric input field set to "1".
- Current Level (1):** A numeric input field set to "1".
- Auto Range (1: ON):** A dropdown menu showing "ON" with a value of "1".
- Range (0: 0):** A numeric input field set to "0".
- Average (0):** A numeric input field set to "0".
- Speed (0: SLOW):** A dropdown menu showing "SLOW" with a value of "0".
- Auto LCZ (T: Enable):** A dropdown menu showing "Enable".
- ALC (F: Disable):** A dropdown menu showing "Disable".
- DC Bias:** A dropdown menu showing "OFF".
- DC Bias Voltage (0):** A numeric input field set to "0".
- Monitor1:** A dropdown menu showing "OFF" with a value of "0".
- Monitor2:** A dropdown menu showing "OFF" with a value of "0".

At the bottom of the interface, there are four monitoring value displays arranged in a 2x2 grid:

- Primary Value:** A display showing "0".
- Secondary Value:** A display showing "0".
- Monitor1 Value:** A display showing "0".
- Monitor2 Value:** A display showing "0".

A small green indicator light is visible between the Secondary Value and Monitor2 Value displays.



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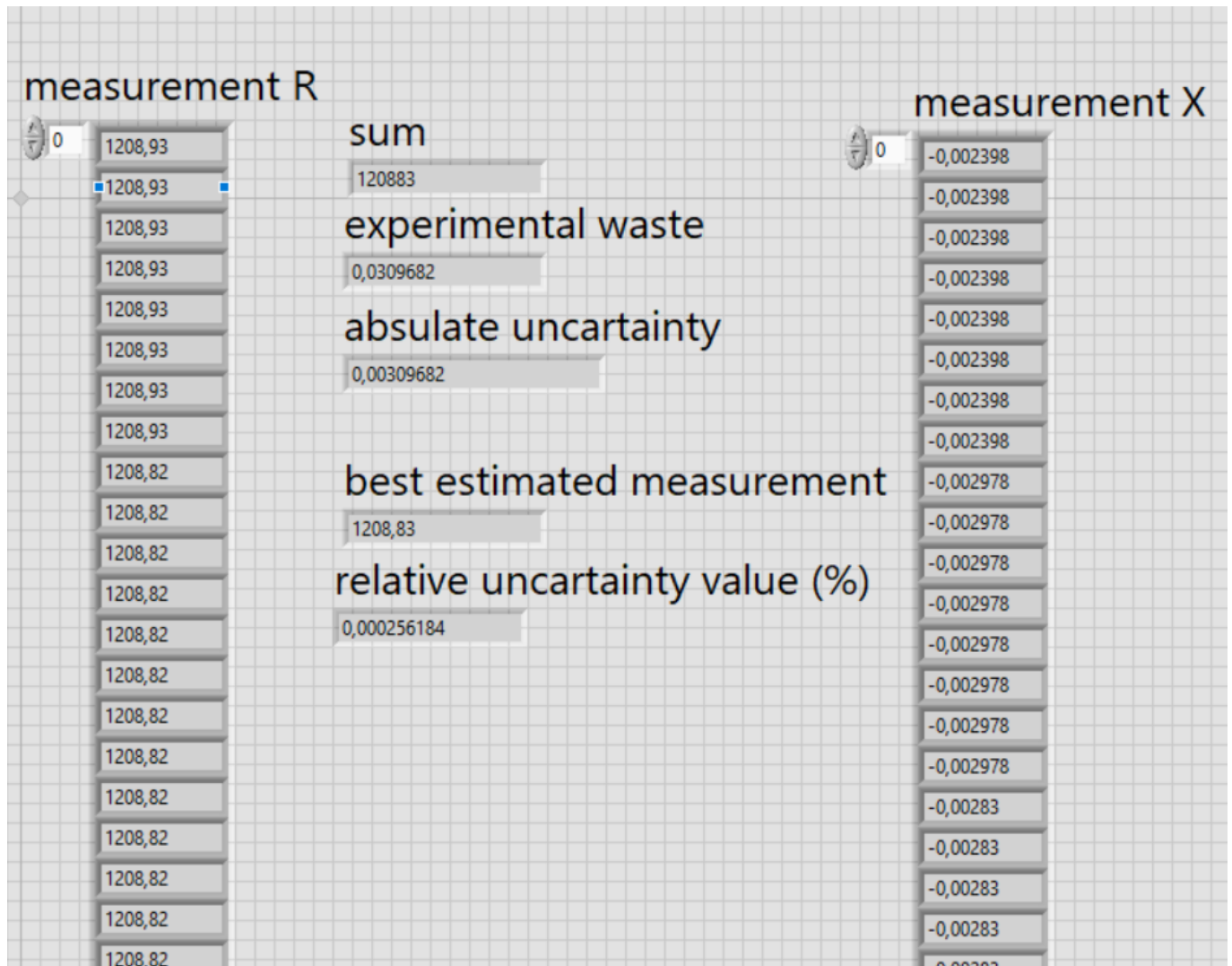
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Resistor 50 hz R-X





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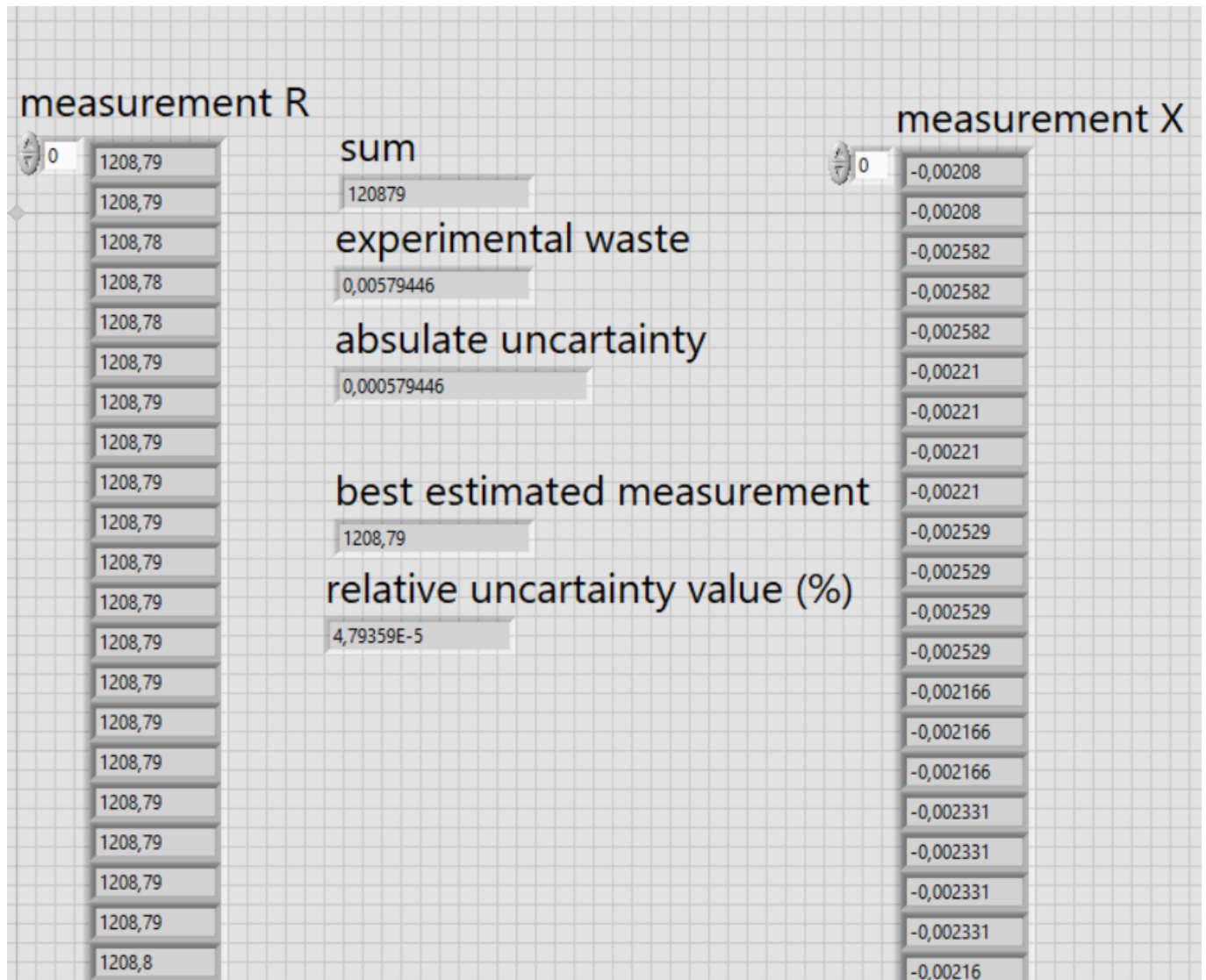
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Resistor 10k hz R-X





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Resistor 300k hz R-X

measurement R		measurement X
0	sum	0
1201,62	120160	-0,110424
1201,62	experimental waste	-0,110424
1201,61	0,00803779	-0,109612
1201,61	absulate uncertainty	-0,109612
1201,61		-0,109612
1201,6		-0,109361
1201,6		-0,109361
1201,6		-0,109361
1201,6	best estimated measurement	-0,109361
1201,61	1201,6	-0,109315
1201,61	relative uncertainty value (%)	-0,109315
1201,61	6,68924E-5	-0,109315
1201,61		-0,109315
1201,6		-0,110035
1201,6		-0,110035
1201,6		-0,110035
1201,61		-0,109066
1201,61		-0,109066
1201,61		-0,109066
1201,61		-0,109066
1201,6		-0,109615



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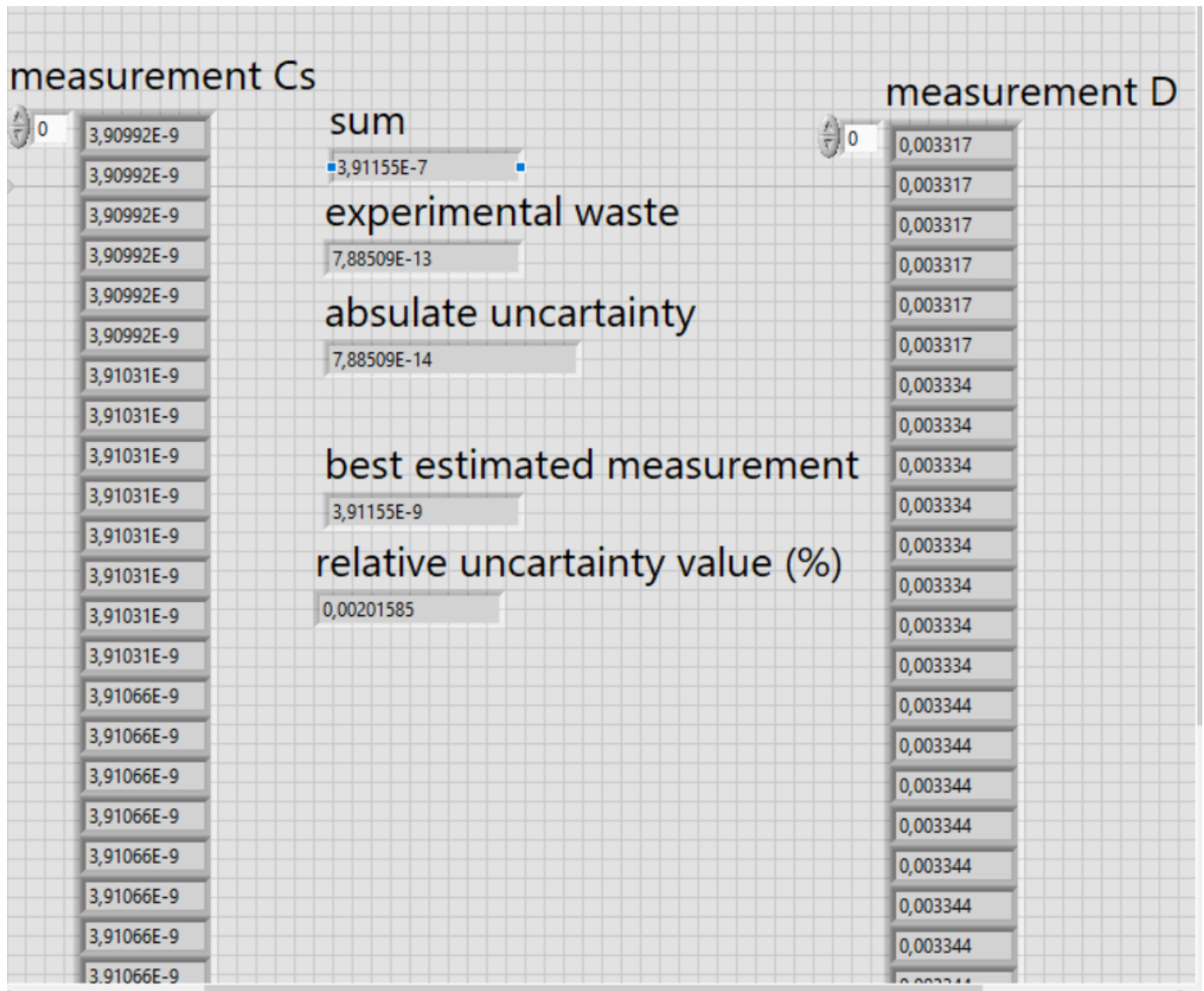
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non-polarized capacitance 50 hz Cs-D





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non-polarized capacitance 10k hz Cs-D

measurement Cs			measurement D	
0	3,84469E-9	sum	0	0,007384
	3,84469E-9	3,84472E-7		0,007384
	3,84469E-9	experimental waste		0,007384
	3,84469E-9	1,60378E-14		0,007384
	3,84469E-9	absolute uncertainty		0,007378
	3,84469E-9	1,60378E-15		0,007378
	3,84474E-9			0,007378
	3,84474E-9	best estimated measurement		0,007381
	3,84474E-9	3,84472E-9		0,007381
	3,84474E-9	relative uncertainty value (%)		0,007381
	3,84472E-9	4,17139E-5		0,007379
	3,84472E-9			0,007379
	3,84472E-9			0,007379
	3,84472E-9			0,007379
	3,84472E-9			0,007379
	3,84472E-9			0,007384
	3,84472E-9			0,007384
	3,84472E-9			0,007384
	3,84469E-9			0,00738
	3,84469E-9			0,00738
	3,84469E-9			0,00738



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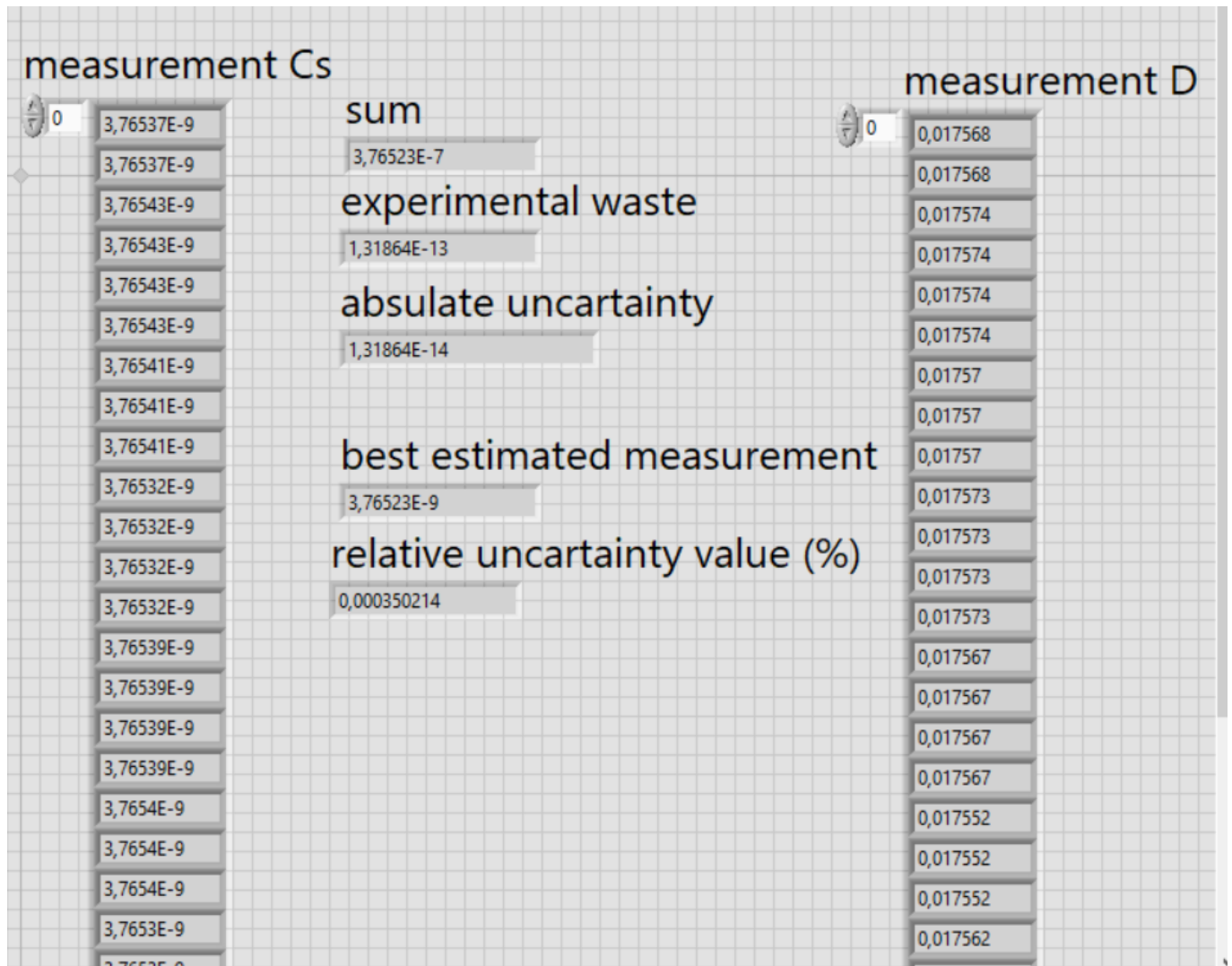
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non-polarized capacitance 300k hz Cs-D





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electrolyte capacitance 10k hz Cs-D

measurement Cs			measurement D	
0	4,1088E-5	sum	0	1,37644
	4,1088E-5	0,00410873		1,37644
	4,10849E-5	experimental waste		1,37642
	4,10849E-5	1,90821E-9		1,37642
	4,10849E-5	abslute uncertainty		1,37642
	4,10849E-5	1,90821E-10		1,37642
	4,10875E-5			1,37636
	4,10875E-5			1,37636
	4,10875E-5	best estimated measurement		1,37636
	4,10891E-5	4,10873E-5		1,37652
	4,10891E-5	relative uncertainty value (%)		1,37652
	4,10891E-5	0,000464428		1,37652
	4,10891E-5			1,37652
	4,10853E-5			1,37638
	4,10853E-5			1,37638
	4,10853E-5			1,37638
	4,10853E-5			1,37638
	4,10886E-5			1,37648
	4,10886E-5			1,37648
	4,10886E-5			1,37648
	4,10897E-5			1,37654
	4,10897E-5			1,37654



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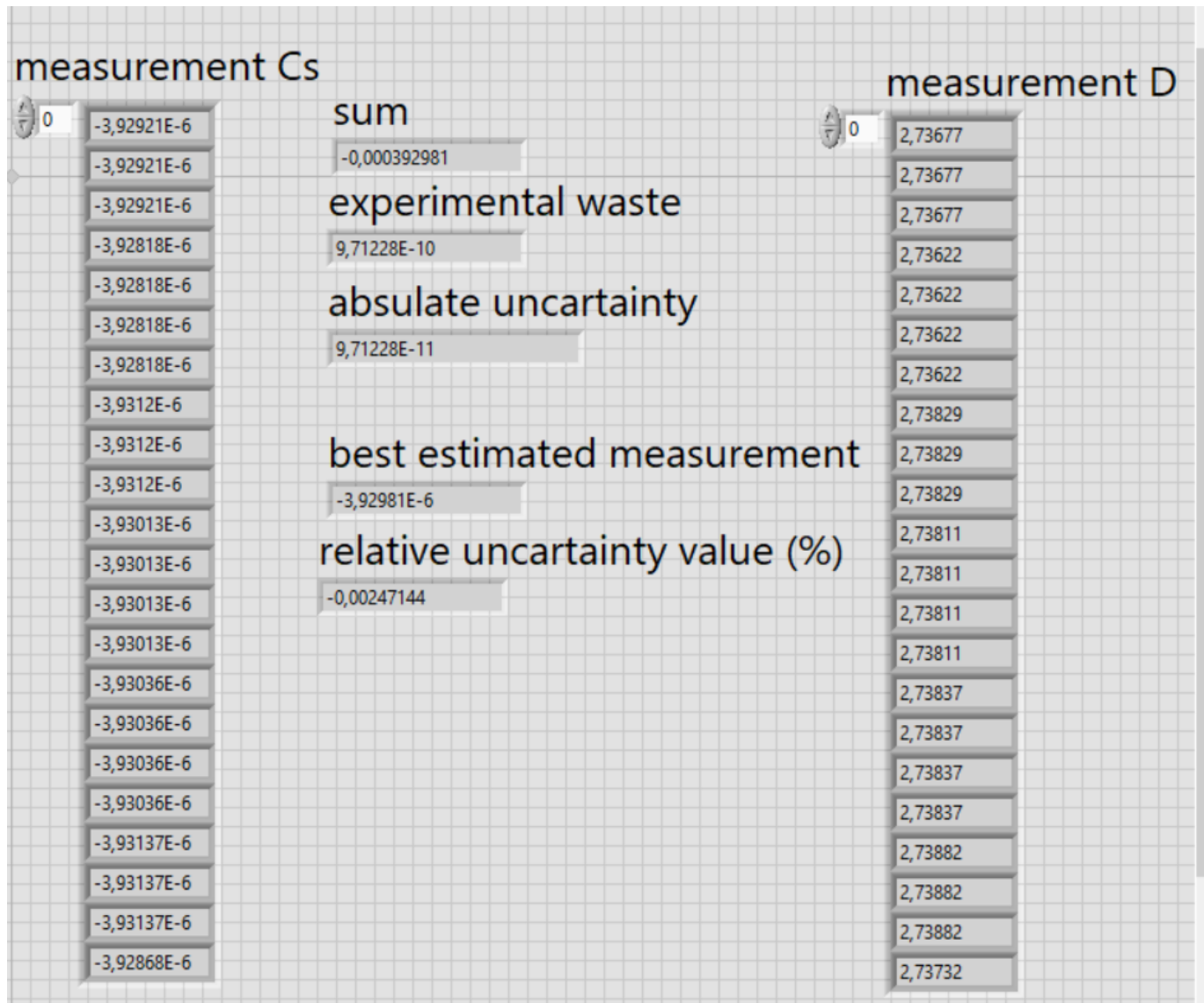


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electrolyte capacitance 300k hz Cs-D





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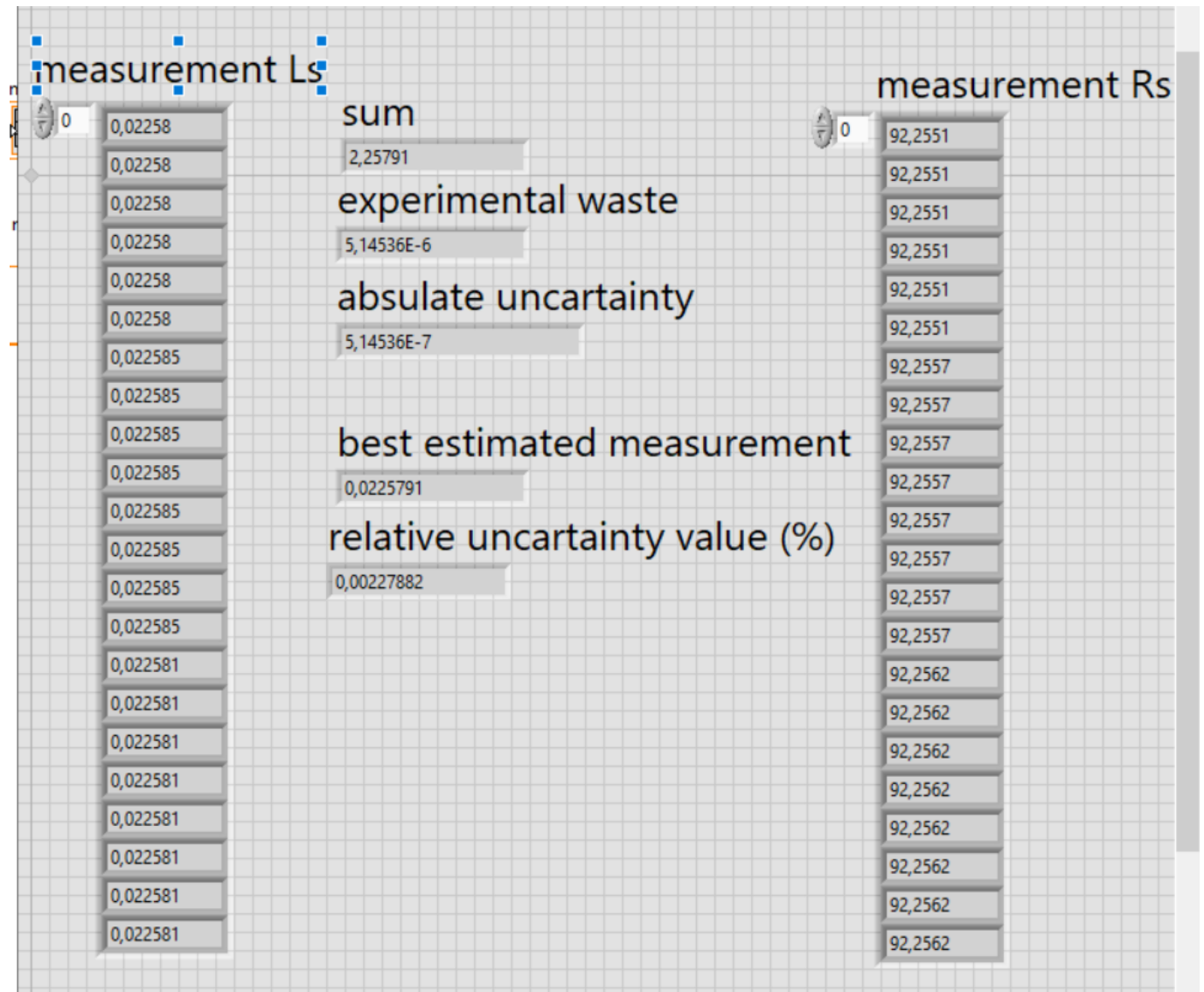
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22mH inductance 50 hz Ls-Rs





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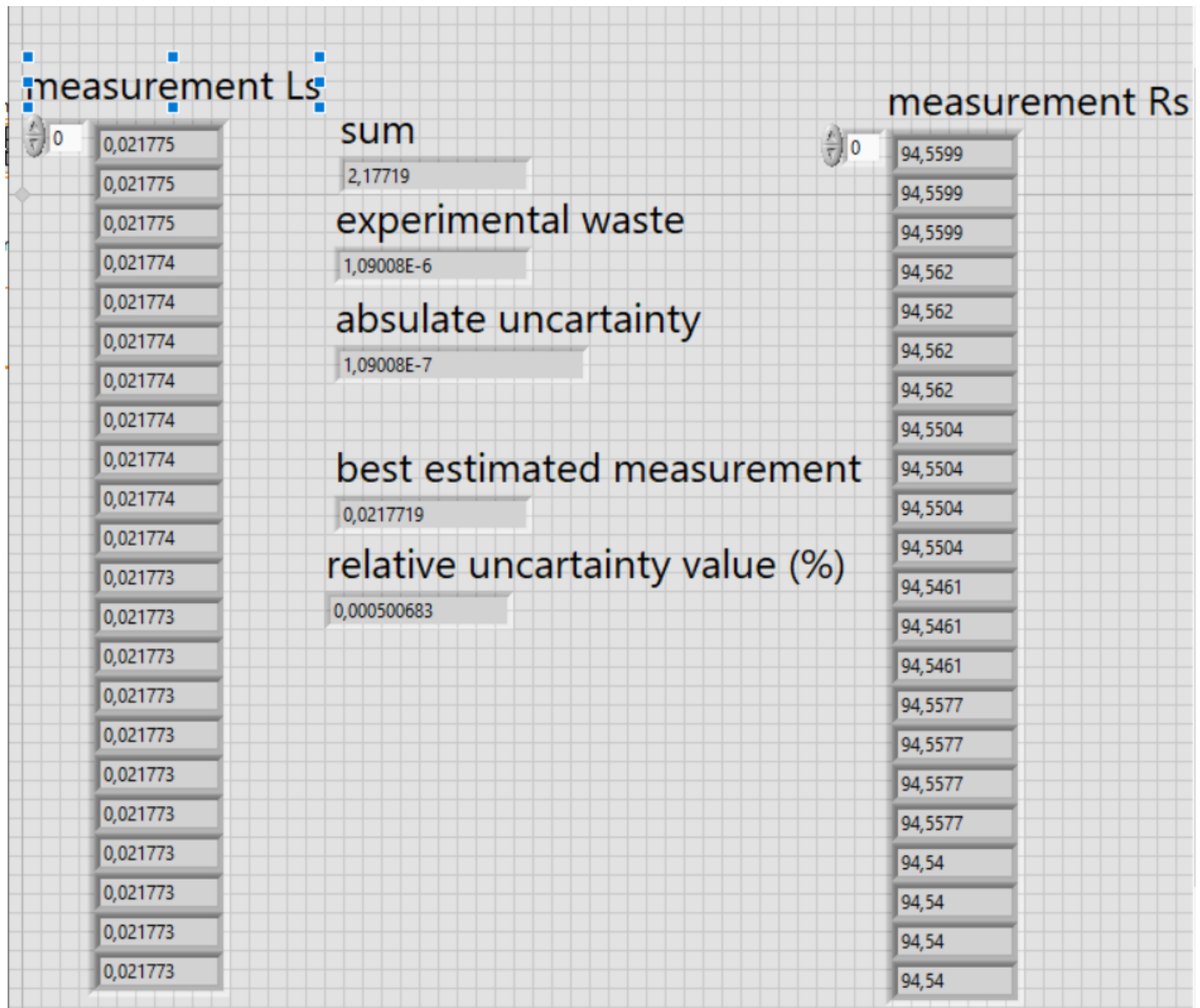
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22mH inductance 10k hz Ls-Rs





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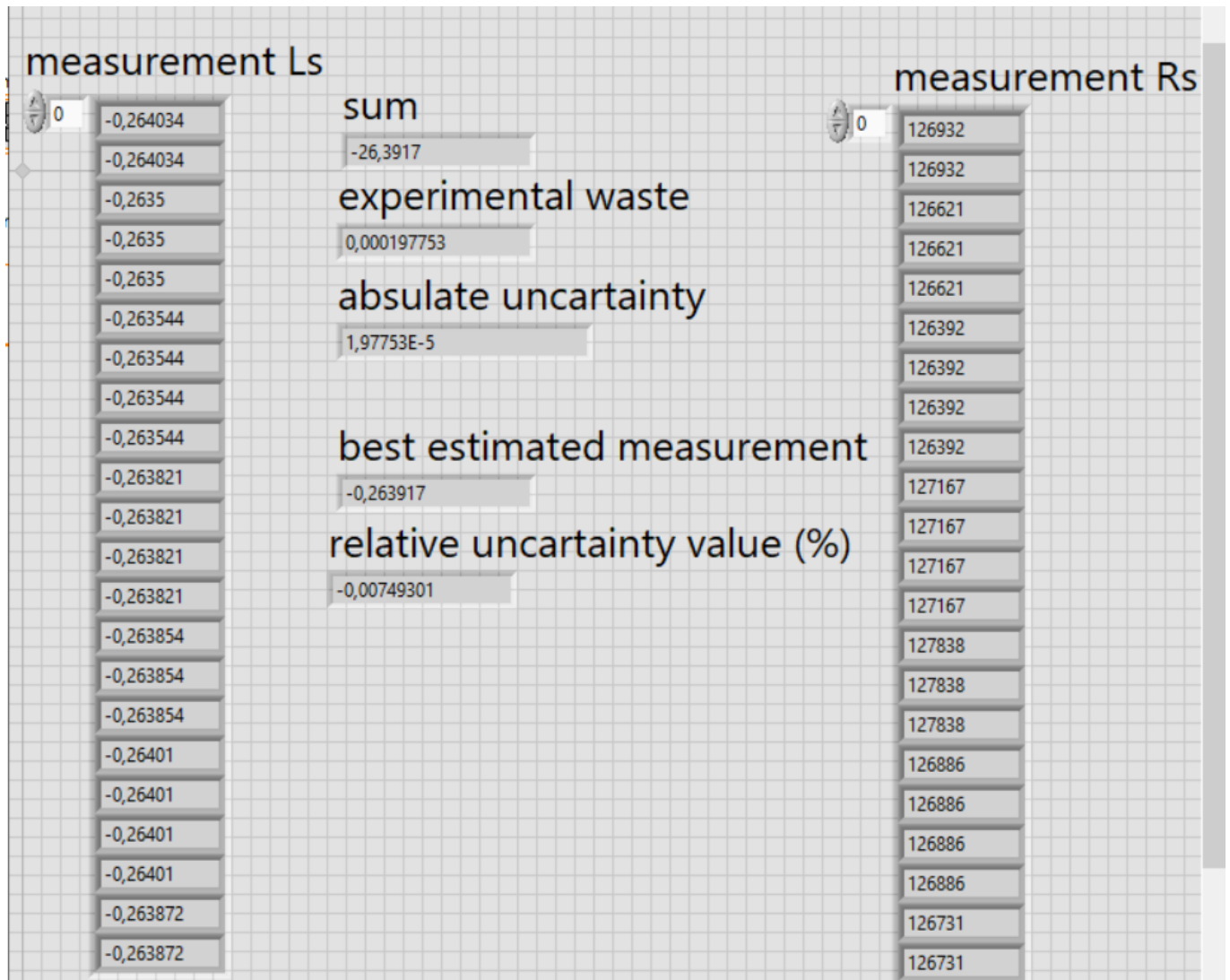
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22mH inductance 300k hz Ls-Rs





Type B uncertainty:

Lcr Meter Datasheet

Measurement speed	Measuring frequency	K_a	K_b
Slow Medium	$f_m < 100\text{Hz}$	$(\frac{1 \times 10^{-3}}{ Z_m })(1 + \frac{200}{V_s})(1 + \sqrt{\frac{100}{f_m}})$	$ Z_m (1 \times 10^{-9})(1 + \frac{70}{V_s})(1 + \sqrt{\frac{100}{f_m}})$
	$100\text{Hz} \leq f_m \leq 100\text{kHz}$	$(\frac{1 \times 10^{-3}}{ Z_m })(1 + \frac{200}{V_s})$	$ Z_m (1 \times 10^{-9})(1 + \frac{70}{V_s})$
	$f_m > 100\text{kHz}$	$(\frac{1 \times 10^{-3}}{ Z_m })(2 + \frac{200}{V_s})$	$ Z_m (3 \times 10^{-9})(1 + \frac{70}{V_s})$
Fast	$f_m < 100\text{Hz}$	$(\frac{2.5 \times 10^{-3}}{ Z_m })(1 + \frac{400}{V_s})(1 + \sqrt{\frac{100}{f_m}})$	$ Z_m (2 \times 10^{-9})(1 + \frac{100}{V_s})(1 + \sqrt{\frac{100}{f_m}})$
	$100\text{Hz} \leq f_m \leq 100\text{kHz}$	$(\frac{2.5 \times 10^{-3}}{ Z_m })(1 + \frac{400}{V_s})$	$ Z_m (2 \times 10^{-9})(1 + \frac{100}{V_s})$
	$f_m > 100\text{kHz}$	$(\frac{2.5 \times 10^{-3}}{ Z_m })(2 + \frac{400}{V_s})$	$ Z_m (6 \times 10^{-9})(1 + \frac{100}{V_s})$

The accuracies of L, C, R, $|Z|$ is equal to A_e , which is defined by:

$$\longrightarrow A_e = \pm [A \times A_r + (K_a + K_b + K_c) \times 100 + K_L] \times K_e \quad [\%]$$

A : Basic measurement accuracy

A_r : Basic accuracy correction factor

K_a : Impedance factor a

K_b : Impedance factor b

K_c : Temperature factor

K_f : Open/short trimming factor

K_L : Test leads' length factor



RESISTANCE	u_R	R	X	u_X
50 Hz	2,0017	(47 ± 2)	$-0,0029 \pm 0,0001$	$-0,000129557$
10 KHz	0,844784	$(47,0 \pm 0,1)$	$0,00456 \pm 0,00008$	$8,33815E - 5$
300 KHz	0,846152	$(46,9 \pm 0,1)$	$0,108 \pm 0,002$	0,00198851

CAPACITOR(Np)	u_C	C	R	u_R
50 Hz	$1,25125E - 9$	$(3,957 \pm 0,001) * 10^{-6}$	$(9,28 \pm 2,94) * 10^7$	$2,93572E + 7$
10 KHz	$4,83157E - 12$	$(3,887 \pm 0,004) * 10^{-9}$	509267 ± 632	632,896
300 KHz	$2,47134E - 11$	$(3,79 \pm 0,02) * 10^{-9}$	$7472,12 \pm 48,63$	48,6381

CAPACITOR(P)	u_C	C	R	u_R
50 Hz	$4,08073E - 9$	$(4,883 \pm 0,004) * 10^{-6}$	$48617,6 \pm 40,6$	40,6197
10 KHz	$1,01929E - 6$	$(4,691 \pm 0,001) * 10^{-3}$	$1,8 \pm 0,4$	0,395674
(L) 300 KHz	$1,03295E - 8$	$(1,86 \pm 0,01) * 10^{-6}$	$1,37 \pm 0,75$	0,755989

INDUCTOR	u_L	L	R	u_R
50 Hz	0,000506186	$(0,0224 \pm 0,0005)$	$89,43 \pm 2,02$	2,01825
10 KHz	$1,74768E - 5$	$(0,02 \pm 0,01) * 10^{-3}$	$92,15 \pm 0,08$	0,0739887
300 KHz	0,0338536	$(0,1898 \pm 0,0001)$	61639 ± 11005	11005,4

Exercise 6:

Measure a nominal 22 mH inductance at 50 Hz using a bridge method. To estimate the parasitic resistance, consider the measurements taken with the LCR meter. Follow the following instructions:

1. Choose the type of bridge among those studied
2. Use a digital multimeter in AC voltage mode as a 0 detector
3. Use a waveform generator that generates a sine wave set to 50 Hz and 1 Vrms.
4. Evaluate the B category uncertainty of the measurement using single or double weighing.
5. CX Compare the uncertainty with that calculated with the LCR meter and evaluate whether the uncertainties are compatible. Make considerations on the comparison of uncertainty with the LCR meter.



Task 1: Measurement of a 22 mH Inductance Using the Maxwell Bridge Method

Instrumentation

- Function Generator
- Dual Measurement Multimeter
- Resistance Decade Boxes (3)
- Capacitor
- Inductor
- Connecting Probes

Objective

- To measure a nominal 22 mH inductance using the Maxwell bridge method, estimate its parasitic resistance, evaluate the measurement uncertainty, and compare the results with those obtained using an LCR meter.

Methodology

- A Maxwell bridge circuit was assembled using the available passive components and resistance decade boxes.
- A sinusoidal excitation signal was generated using the function generator at **50 Hz** with an amplitude of **1 V_{rms}**.
- A digital multimeter operating in AC voltage mode was used as a **null detector (0 detector)** to identify the bridge balance condition.
- The bridge components were adjusted until the detector indicated a minimum voltage, corresponding to the balanced state of the bridge.
- The inductance value and parasitic resistance of the unknown inductor were calculated using the Maxwell bridge balance equations.
- Category B uncertainty was evaluated using the specifications and tolerances of the bridge components and measuring instruments.
- The same inductor was subsequently measured using an LCR meter, and the obtained inductance and parasitic resistance values were recorded.
- The bridge results were compared with the LCR meter measurements to assess measurement compatibility.

$d\lambda$ is the resolution of the multimeter = 0,01mV;

- ΔR_X is the minimum variation of $R_1 = 1\text{ohm}$;
- R_x is the balancing value = 108 ohm;
- $\Delta \lambda$ is the variation of the voltage measured by the DMM given by $\Delta R_x = 1,52\text{ mV}$



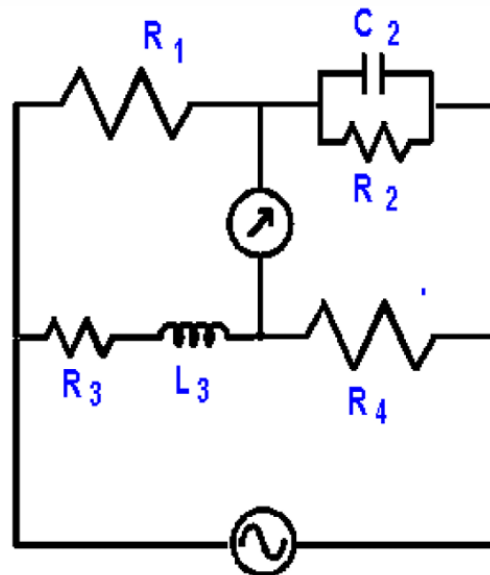
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$$R_2 = \frac{R_1 \times R_4}{R_3}$$

$$R_3 = \frac{100 \times 100}{108} = 92,5929$$

$$L_3 = R_1 \times R_4 \times C_2 = \frac{22,51}{0,02251} \text{ mH}$$

$$V_L(L) = \sqrt{V_L(R_1)^2 + V_L(R_4)^2 + V_L(C_2)^2 + V_0^2}$$

$$= \sqrt{0,577^2 + 0,577^2 + 1,155^2 + (0,000356)^2}$$

$$V_L(L) = 1,41421\%$$

$$V_L(R_x) = \sqrt{V_L(R_1)^2 + V_L(R_2)^2 + V_L(R_4)^2 + V_0^2}$$

$$= \sqrt{1} = 1\%$$

$$V(R_x) = 92,59 \times \frac{1}{100} = 0,93$$

$$V(L) = 22,51 \times 10^{-3} \times \frac{1,41}{100} = 0,318 \text{ mH}$$

$$R_x = (92,59 \pm 0,93) \Omega \quad L = (22,51 \pm 0,318) \text{ mH}$$

$R_1 = 100$
 $R_4 = 100$
 $R_2 = 108$
 $C_2 = 2,251 \text{ mF}$
 $C_2 = 0,2 \text{ accuracy}$
 $R_1, R_2, R_4 = 0,1 \text{ accuracy}$

$V_L(R) = \frac{0,1}{\sqrt{2}} = 0,577\%$
 $V_L(C) = \frac{0,2}{\sqrt{2}} = 1,155\%$

$V_0 = \frac{\Delta R_x}{R_x} \cdot \frac{1}{100\sqrt{2}}$
 $V_0 = \frac{0,01 \times 10^{-3} \times 1}{108} \cdot \frac{1}{1,52 \times 10^{-2} \times 100\sqrt{2}}$
 $V_0 = 0,000356\%$



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Comparison of the Results and Uncertainties

Resistance Comparison

Bridge measurement:

$$R_x = (92.59 \pm 0.93) \, \Omega$$

LCR meter measurement:

$$R_s = (92.164 \pm 0.306) \, \Omega$$

The absolute difference between the two measurements is:

$$\Delta R = |R_x - R_s|$$

$$\Delta R = |92.59 - 92.164|$$

$$\Delta R = 0.426 \, \Omega$$

The combined standard uncertainty is:

$$u_c(R) = \sqrt{u^2(R_x) + u^2(R_s)}$$

$$u_c(R) = \sqrt{0.465^2 + 0.306^2}$$

$$u_c(R) = 0.557 \, \Omega$$



Inductance Comparison

Bridge measurement:

$$L = (22.51 \pm 0.318) \text{ mH}$$

LCR meter measurement:

$$L_s = (22.58 \pm 0.075) \text{ mH}$$

The absolute difference between the measurements is:

$$\Delta L = |L - L_s|$$

$$\Delta L = |22.51 - 22.58|$$

$$\Delta L = 0.07 \text{ mH}$$

The combined standard uncertainty is:

$$u_c(L) = \sqrt{u^2(L) + u^2(L_s)}$$

$$u_c(L) = \sqrt{0.318^2 + 0.075^2}$$

$$u_c(L) = 0.327 \text{ mH}$$

The values obtained with the bridge method and with the GW Instek LCR-6300 are compatible within their respective uncertainty intervals. For the resistance measurement, the difference between the two methods is $0.426 \, \Omega$, which is lower than the combined standard uncertainty ($0.557 \, \Omega$). For the inductance measurement, the difference is only 0.07 mH compared with a combined uncertainty of 0.327 mH . Since $En < 11$ in both cases, the bridge and LCR meter measurements can be considered mutually consistent. The LCR meter provides a lower uncertainty, while the bridge method yields results in agreement with the reference instrument.